

HP Smart Update Manager User Guide

Version 6.0.0

Abstract

This document describes how to use HP SUM to apply firmware updates to HP ProLiant and HP Integrity servers, and apply software updates to HP ProLiant servers. This document is intended for individuals who understand the configuration and operations of Microsoft Windows, Windows Server, Linux, smart components, HP-UX, VMware, and the risk of data loss from performing updates.



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1 Introduction

HP SUM overview

HP SUM is a technology included in many HP products for installing and updating firmware and software on HP ProLiant servers, and firmware on HP Integrity servers.

HP SUM provides a web-based GUI and a command-line scriptable interface for:

- Deployment of firmware for single or one-to-many HP ProLiant and HP Integrity servers and network-based targets such as iLO, OA, and VC Ethernet and Fibre Channel modules.
- Deployment of software for single or one-to-many HP ProLiant servers (supported in Windows and Linux environments).

HP SUM has an integrated hardware and software discovery engine that finds the installed hardware and current versions of firmware and software in use on nodes you identify. HP SUM installs updates in the correct order and ensures that all dependencies are met before deploying an update. HP SUM prevents an installation if there are version-based dependencies that it cannot resolve.

Key features of HP SUM include:

- Dependency checking, which ensures appropriate installation order and component readiness
- Automatic and wizard-like Guided Update process
- Web browser based application
- Create custom baselines and ISOs
- Assign specific baselines to nodes
- Download updates from the web
- Intelligent deployment of only required updates
- Simultaneous firmware and software deployment for multiple remote nodes in GUI and CLI modes
- Improved deployment performance
- Local online deployment of HP ProLiant servers and enclosures
- Remote (one-to-many) online deployment of HP ProLiant and HP Integrity servers and enclosures
- Local offline firmware deployments with HP Service Pack for ProLiant deliverables
- Remote offline deployment when used with the SmartStart Scripting Toolkit (HP ProLiant G7 and earlier servers), Scripting Toolkit (HP ProLiant Gen8 and later), iLO Virtual Media, or PXE booted media
- GUI or CLI scripts with extensive logging
- Remote command-line deployment
- Support for updating firmware on network-based targets such as the OA, iLO through the Network Management Port, VC Ethernet and Fibre Channel switches, and 3Gb/6Gb SAS BL Switch interconnects on HP ProLiant servers
- Support for deploying firmware updates to supported Integrity servers and Superdome 2 enclosures
- Support for updating VC modules on Integrity servers

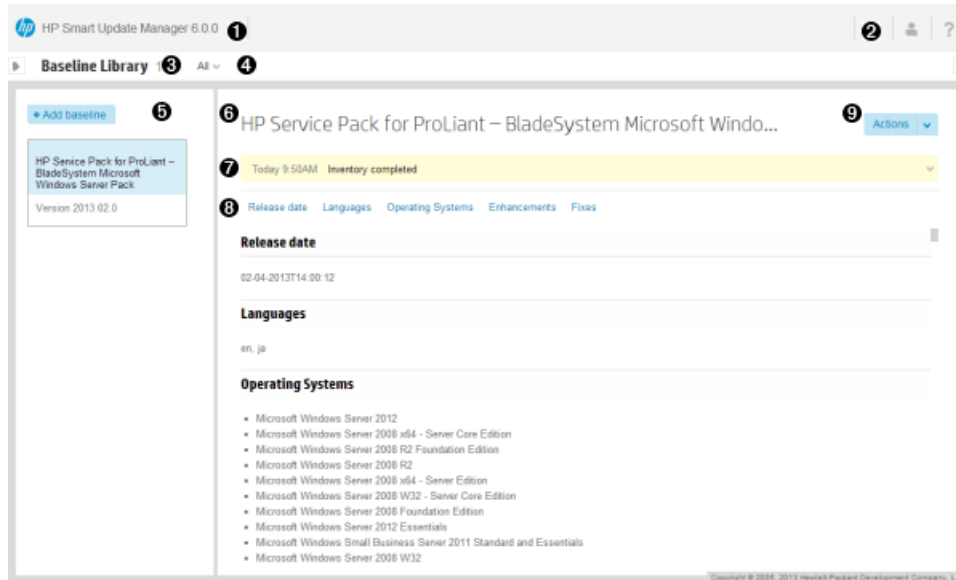
NOTE: HP SUM does not support third-party controllers. This includes flashing hard drives behind these controllers.

- Remote online deployment of I/O Card firmware on HP ProLiant and HP Integrity targets running HP-UX

About the graphical user interface

HP SUM is a browser-based application. The screenshot below shows the major sections of the screen.

NOTE: Do not use the **Back** button press **F5** in your browser window.



Item	Description
1	Main menu - This menu provides links to all HP SUM screens.
2	• Activity list - Status updates appear briefly, and the activity icon displays the status of recent activities. Click the Activities icon, to open the Activity panel. • Login information - Displays the currently logged-in user, and offers a logout function. • Help panel - Opens the help pane.
3	Screen name - Displays the name of the screen.
4	Screen filters - Filters screen objects.
5	Screen list - Displays a list of items on this screen.
6	Baseline or Node name - This lists the baseline or node that you selected in the screen list. This example shows the name of the selected baseline.
7	Informational highlight - This bar highlights information that you might need to perform or know about for the selected item. Click to expand this box if there is more information.
8	Item details - This area provides full details about the selected item.
9	Actions - Lists the available actions. NOTE: Screen options vary based on the screen you are viewing.

Status icon descriptions

HP SUM uses icons to represent the current status of resources and alerts and to control the display.

Table 1 Status icons

Large icon	Small icon	Description
		Critical Failed/Interrupted
		Warning
		OK Successful
		Disabled
		Unknown
		An In progress rotating icon indicates a change is being applied or a task is running.

HP SUM new terms

The following terminology has changed from HP SUM 5.x to HP SUM 6:

HP SUM 5.x	HP SUM 6	Description
Repository	Baseline	A collection of components for use in updating a node. For example, HP SPP or a custom baseline.
Target	Node	Hardware you want to update.
Discovery	Scouting	The process where HP SUM determines node type.
Discovery	Inventory	The process where HP SUM determines the contents of a node or baseline.
N/A	Location	Network directory for baseline updates.
N/A	Guided Update	An automated or wizard-style method to update the local node.
N/A	Mode	You can run HP SUM in either interactive or non-interactive mode.
N/A	Analyze	The process where HP SUM verifies all dependencies are met before beginning deployment.
Install	Deploy	The process where HP SUM begins the update process.

2 Downloading, installing, and launching HP SUM

Downloading HP SUM

Downloading HP SUM from the HP SUM website

You can download the latest version of HP SUM from the HP SUM website at <http://www.hp.com/go/hpsum>.

Unzip the file and save it to a directory on your system.

NOTE: Do not put HP SUM 6.x files in the same directory as HP SUM 5.3.5 and earlier versions of HP SUM.

Downloading HP SUM from the SDR website

You can download HP SUM as an RPM from the HP Software Delivery Repository at <http://downloads.linux.hp.com/>. The SDR contains a version of the HP SUM RPM for each supported operating system and architecture type.

You can use yum commands to search for HP SUM and download it to your system. You can also use a web browser to navigate the HP SDR and download the rpm.

Use the following commands to search, download, or install HP SUM from the SDR:

Action	Command example
Search for HP SUM with yum	<code>yum search hpsum</code>
Install HP SUM with yum	<code>yum install hpsum</code>
Download HP SUM from the SDR with a web browser, and then install the RPM.	<code>rpm -Uvh hpsum-5.3.5-66.rhwl5.i386.rpm</code>

For more information on using the HP SDR, see the Getting Started and FAQ sections on the HP Software Delivery Repository website at <http://downloads.linux.hp.com/SDR/index.html>.

For more information on using HP SUM with the SDR, see *Linux best practices using HP Service Pack for ProLiant (SPP) and Software Delivery Repository (SDR)* at <http://h20000.www2.hp.com/bc/docs/support/SupportManual/c03479393/c03479393.pdf>.

Migrating nodes from HP SUM 5.x to HP SUM 6

HP provides scripts that migrate the targets you added to HP SUM 5.x into nodes that HP SUM 6 recognizes. If you want to edit the targets that you migrate to HP SUM 6, use the `/script` parameter. This generates a file that you can edit with a text editor application.

NOTE: The migration script does not support sudo credentials.

To migrate nodes to HP SUM 6:

1. Navigate to the directory that holds HP SUM.
2. Double-click `port_targets.bat` (Windows) or `port_targets.sh` (Linux).

You can also run the scripts from the command line.

Script syntax:

```
port_targets.bat [/y] [/h] [/script] [/user {username} /password {password}]
```

The script supports the following options:

Option	Definition
/y	Continue the migration without using prompts.
/h	Print the script help.
/script	This option creates a file with the migration commands, it does not run the script. You can edit the script commands manually before migrating the targets. NOTE: Use this option to select the targets you migrate, or edit the user credentials for an individual target.
/user {username} /password {password}	This options sets a default username and password credentials for each node that HP SUM migrates. If you do not use this option, each node is given the following credentials: • user=username • password=passwd

Deploying HP SUM

The following sections discuss deployment from HP SUM. For more information on planning an update, see the *HP ProLiant and Integrity Firmware Management Best Practices Overview*, *HP ProLiant and Integrity Firmware Management Best Practices Planning Guide*, or *HP ProLiant and Integrity Firmware Management Best Practices Implementer Guide*, available on the HP website:

<http://www.hp.com/go/hpsum/documentation>.

Running HP SUM

HP SUM provides the following modes:

- GUI—For firmware and software deployments
- Scripted—For silent firmware and software deployments

HP SUM supports local and remote deployments. HP SUM runs in Windows and Linux in online and offline mode. For more information about deployment modes, see “[Deployment modes](#)” (page 12).

NOTE: You cannot use Windows Hyper-V systems to run HP SUM and deploy updates. You can select Hyper-V systems as a node from HP SUM running on a Windows or Linux system.

NOTE: Before deploying software updates to a server, be sure that a recent backup of the server is available in the event the deployment procedure fails.

For information on the minimum requirements to run HP SUM, see the *HP Smart Update Manager Release Notes* on the HP website:

<http://www.hp.com/go/hpsum/documentation>

Deploying firmware for HP ProLiant servers using the HP Service Pack for ProLiant

The SPP is a re-packaging of HP ProLiant system software and firmware for HP ProLiant BL/ML/DL/SL servers and their options, and BladeSystem enclosures including OA, VC, and 3 Gb SAS switches running supported Windows, Linux, VMware (supported firmware updates only) operating systems. The single SPP image contains a comprehensive collection of firmware and system software components including drivers, agents, tools, utilities, and firmware that is tested, managed, and deployed together as a single solution.

The SPP includes HP SUM for deployment of the SPP components.

For a complete list of HP ProLiant systems and software supported by SPP, and updates available in an SPP release, see the *HP Service Pack for ProLiant Release Notes* on the HP website:

<http://www.hp.com/go/spp/documentation>

Deploying HP Integrity firmware bundles with HP SUM

You can use HP SUM to deploy components delivered with the HP Integrity firmware bundles.

Deploying firmware to HP Integrity servers is done remotely and is the same as deploying other firmware, such as OA, with the following exceptions:

- Specify the IP address of the Monarch (primary) OA on Superdome 2, or the Monarch iLO 3 management processor on multi-blade servers.
- Specify the server IP address as the target to update I/O firmware on HP-UX servers. You can only update remote HP-UX targets.

For more information on HP Integrity firmware bundles, see the Manage HP Integrity Servers Firmware Updates website at <http://www.hp.com/go/smartupdate/integrity>.

Deployment modes

The following key terms apply when using HP SUM to deploy updates:

Term	Definition
Local	The installation runs on the physical hardware you are updating. For example, running a utility on a server to update the system ROM of the server.
Remote	The installation runs on one system, but updates other physical nodes. For example, updating the OA or HP Integrity server firmware across a network.
Online	The installation occurs while the host processor is running in the normal server environment. For example, if the server runs Microsoft Windows Server 2012, the update occurs under this environment. The update does not require you to boot to a special environment to update the firmware. You might need to reboot the node to activate the firmware.
Offline	In offline mode, the HP SUM boots a small Linux kernel and enables updates to occur on a single server. • Only updates the local system • Only uses a single baseline NOTE: Some features of HP SUM that require the regular local host operating systems are not supported in offline mode.

These terms can be used in combination to designate the type of environment required for updates to occur, such as local-online or remote-online.

Deployment scenarios

HP SUM deploys updates from a local host to one or more remote hosts. If the host running HP SUM uses Windows, you can update Windows, Linux, VMware, or HP-UX targets. If the host running HP SUM uses Linux, you can update Linux, VMware, or HP-UX nodes. You can also update remote HP ProLiant or HP Integrity iLO, OA, and VC nodes from Windows or Linux systems.

The following table describes when typical HP SUM deployment scenarios are used.

Scenario	Used when
Graphical deployment on a local host	• You are not familiar with command line tools. • You are deploying components on a local, single host. • Updates do not require scripting.
Scripted deployment on a local host	• You are familiar with command line tools. • You are deploying components on a local, single host. • Updates require a customized, scripted deployment.
Graphical deployment to a remote host	• You are not familiar with command line tools. • You are deploying components on one or more remote hosts. • Updates do not require scripting.
Scripted deployment to a remote host	• You are familiar with command line tools. • You are deploying components on one or more hosts. • Updates require a customized, scripted deployment to one or more host systems.

Disabling BitLocker to permit firmware updates (Windows only)

The TPM, when used with BitLocker, measures a system state. Upon detection of a changed ROM image, it restricts access to the Windows file system if the user cannot provide the recovery key. HP SUM detects if a TPM is enabled in your system. For some newer models of HP ProLiant servers, if a TPM is detected in your system or with any remote server selected as a target, HP SUM utilities for HP iLO, Smart Array, NIC, and BIOS warn users prior to a flash. If the user does not temporarily disable BitLocker and does not cancel the flash, the BitLocker recovery key is needed to access the user data upon reboot.

A recovery event is triggered in the following situations:

- You do not temporarily disable BitLocker before flashing the system BIOS when using the Microsoft BitLocker Drive Encryption.
- You have optionally selected to measure HP iLO, Smart Array, and NIC firmware.

If HP SUM detects a TPM, a warning message appears:

CAUTION: A Trusted Platform Module (TPM) has been detected in this system. Failure to perform proper OS encryption procedures will result in loss of access to your data if recovery key is not available. Recommended procedure for Microsoft Windows (R) BitLocker (TM) is to \"suspend\" BitLocker prior to System ROM or Option ROM firmware flash. If you do not have your recovery key or have not suspended BitLocker, exit this flash. Failure to follow these instructions will result in loss of access to your data.

To enable firmware updates without the need to type in the TPM password on each server, the BitLocker Drive Encryption must be temporarily disabled. Disabling the BitLocker Drive Encryption keeps the hard drive data encrypted. However, BitLocker uses a plain text decryption key that is stored on the hard drive to read the information. After the firmware updates have been completed,

the BitLocker Drive Encryption can be re-enabled. Once the BitLocker Drive Encryption has been re-enabled, the plain text key is removed and BitLocker secures the drive again.

- △ **CAUTION:** Temporarily disabling BitLocker Drive Encryption can compromise drive security and should only be attempted in a secure environment. If you are unable to provide a secure environment, HP recommends providing the boot password and leaving BitLocker Drive Encryption enabled throughout the firmware update process. This requires setting the `/tpmbypass` parameter for HP SUM or the firmware update is blocked.

To temporarily disable BitLocker support to allow firmware updates:

1. Click **Start**, and then search for `gpedit.msc` in the Search Text box.
2. When the Local Group Policy Editor starts, click **Local Computer Policy**.
3. Click **Computer Configuration**→**Administrative Templates**→**Windows Components**→**BitLocker Drive Encryption**.
4. When the BitLocker settings are displayed, double-click **Control Panel Setup: Enable Advanced startup options**.
5. When the dialog box appears, click **Disable**.
6. Close all windows, and then start the firmware update.

To enable advanced startup options:

1. Enter `cscript manage-bde.wsf -protectors -disable c:`
2. When the firmware update process is completed, the BitLocker Drive Encryption support can be re-enabled by following steps 1 through 4 but clicking **Enabled** in step 5 instead. The following command can be used to re-enable BitLocker Drive Encryption after firmware deployment has completed.
3. Enter `cscript manage-bde.wsf -protectors -enable c:`

The following table describes TPM detection scenarios that you might encounter.

Scenario	Result
If TPM is detected and enabled, the installation is not silent, and a system ROM must be updated.	A warning message appears. Select OK to continue. The installation is not canceled.
If TPM is detected and enabled, the installation is silent, the <code>/tpmbypass</code> switch is not given, and any firmware updated must be applied to the server.	No warning appears. A new log file is generated (%systemdrive%\cpqsystem\log\cpqstub.log). Because the installation is silent, the installation is terminated and cannot continue.
If TPM is detected and enabled with Option ROM Measuring, the installation is not silent, and a system ROM must be updated.	A warning message appears. After selecting OK , you can continue. The installation is not canceled.
If TPM is detected and enabled with Option ROM Measuring, the installation is silent, the <code>/tpmbypass</code> switch is not given, and any firmware updated must be applied to the server.	No warning appears. A new log file is generated (%systemdrive%\cpqsystem\log\cpqstub.log). Because the installation is silent, the installation is terminated and cannot continue.
If TPM is detected and enabled, the installation is silent, the installation occurs, and the <code>/tpmbypass</code> switch is supplied.	The installation occurs.

Launching and logging into HP SUM

Launching HP SUM

HP SUM supports 32-bit and 64-bit processors. When you launch HP SUM, a script chooses the version of HP SUM to run. HP SUM logs you in using your current user credentials. To run HP SUM, your userid needs to be part of the administrator group, or, on a Linux system, you can run HP

SUM using the `sudo` command. If you are using Windows Server 2012, you must log in with the Administrator account, or select **Run as Administrator**.

NOTE: Do not open HP SUM in more than one browser tab or window or run HP SUM CLI commands when the GUI is open.

For more information about supported browsers, see the HP Smart Update Manager Release Notes, available at the HP Smart Update Manager Information Library, <http://www.hp.com/go/hpsum/documentation>.

Launching HP SUM from an HP SUM download on a Windows host

1. Unzip the file you downloaded.
2. In the HP SUM directory, double-click `hpsum.bat`.

Launching HP SUM from an HP SUM download on a Linux host

1. From a command-line, type `./hpsum`.

Logging into HP SUM

If HP SUM is already running, and no user is logged in, use your computer's credentials to log into HP SUM.

Logging out of the HP SUM GUI

1. Click the user icon, and then click **Logout**.
2. Select one of the following:
 - **Log Off - Current user.**
 - **Shutdown - This option will shutdown HP SUM service.**
3. Click **OK**.

HP SUM stores node information between sessions, including user credentials for nodes. The information is stored in a database file in hex format. To clear the information, run the `clean-cache.cmd` (Windows) or `clean-cache.sh` (Linux). To run `clean-cache`:

1. Shut down the HP SUM service.
2. From a command-line window, navigate to the directory that contains HP SUM.
3. Type `clean-cache.bat` (Windows) or `clean-cache.sh` (Linux).

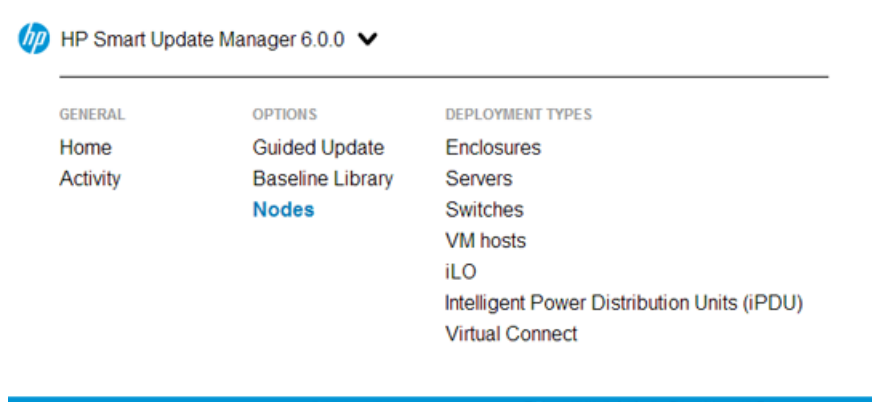
NOTE: If you are using a GUI, you can navigate to the directory that holds HP SUM and double-click the file to clear the cache.

NOTE: Running the `clean-cache` command erases all nodes, baselines, and other information entered in HP SUM.

3 Using the HP SUM GUI

Using the Home screen

When you launch HP SUM, HP SUM displays the Home screen. From this screen, you can click **Guided Update**, **Baseline Library**, or **Nodes**. There is also a **Get Started** button that launches Guided Update. To navigate to other screens, use the navigation menu in the upper-left corner.



Using the Activity screen

The Activity screen provides a brief update of the activities that HP SUM is performing, or has recently performed. The table displays the source, message, state, and last update time of an activity.

Using Guided Update

HP SUM's Guided Update function allows you to update the localhost using the default baseline. The default baseline is the directory from which you are running HP SUM.

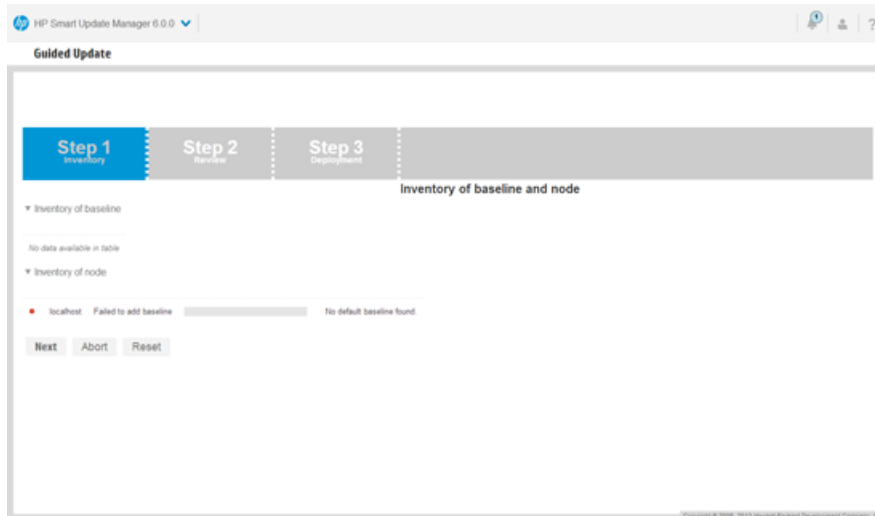
NOTE: If the directory from which you are running HP SUM has no updates, then you cannot use Guided Update. If there is no default baseline, HP SUM displays a message in the Inventory of Baseline section.

- **Interactive:** In this mode HP SUM uses the default baseline to update the localhost. You can choose the updates that HP SUM applies to the localhost.

NOTE: HP SUM does not support configuring components in Guided Update.

- **Automatic:** In this mode HP SUM automatically updates the localhost and uses the default baseline.

NOTE: In automatic mode, HP SUM automatically begins updates when you click OK.



Using Guided Update in Interactive mode

1. In the navigation menu, click **Guided Update**.
2. Click **Interactive**.
3. Click **OK**.
4. HP SUM displays the baseline and the localhost.

NOTE: HP SUM automatically starts to take inventory of the baseline and the node. The inventory process duration varies based on the number of updates in the baseline.

5. Click **Next** after HP SUM finishes the inventory.

Recommended updates ready to deploy are highlighted in blue and HP SUM displays a green icon in the **Ready to proceed** column.

NOTE: Items that are up-to-date or optional are not highlighted. Click **Force** to deploy these updates. If you want to globally select all components to rewrite or downgrade, click **Actions+Advanced Options** and select **Rewrite** or **Downgrade**. Click **Analyze**, and HP SUM performs an analysis again.

NOTE: Double-click an update to view the component details. If you want to view the information in the new update, click the hyperlink.

6. Click **Deploy**.
HP SUM performs component analysis before deploying updates.
7. Review the installation log files.

Using Guided Update in Automatic mode

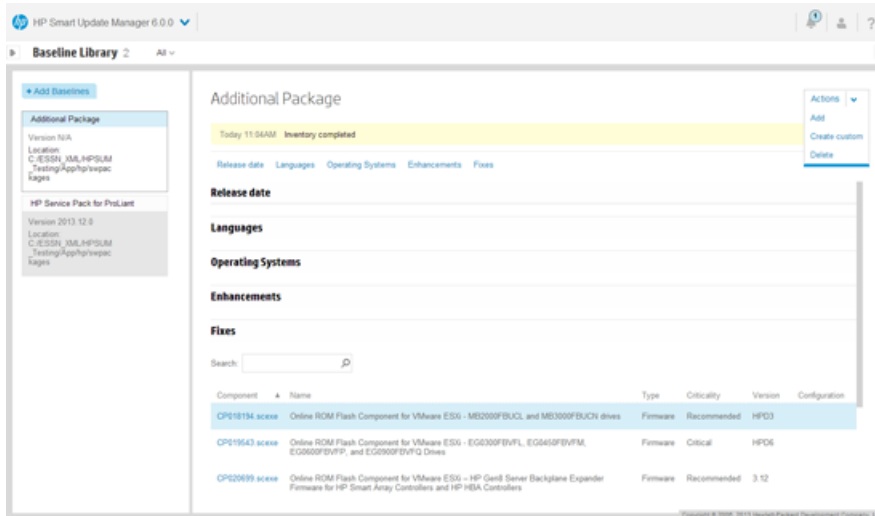
1. In the navigation menu, click **Guided Update**.
2. Click **Automatic**.
3. Click **OK**.

NOTE: HP SUM begins to deploy updates after you click **OK**. HP SUM does not support aborting updates in the Guided Update Automatic mode.

NOTE: HP SUM does not support installing SNMP, WBEM Providers, and AMS components in automatic mode. Use Interactive mode if you want to install these components.

Using the Baseline Library screen

The Baseline Library screen displays the baselines and additional packages you will use to update your nodes. Baselines include the HP SPP, HP Integrity bundle, or custom baselines that include updates that you select from other baselines or additional packages. Additional packages are directories that hold updates that are not included in a named update package, for example a Hot Fix. HP SUM saves baseline information between sessions.



HP SUM displays each added baseline in the left pane. Select a baseline, and HP SUM displays the following information for each baseline and additional package.

- Release date
- Languages
- Operating systems
- Enhancements
- Fixes – HP SUM includes a search box for finding updates that meet the search parameters. You can search on file name, description, and update type. Double-click an update to view its details.
- Component configuration – You need to configure some components before you can deploy the update. For more information about configuring components, see [“Configuring components”](#) (page 21).

Adding a baseline

1. On the Baseline Library screen, click **Add baselines**.
2. Do one of the following:

To add this baseline:	Follow these steps:
HP SUM default location: This location uses a directory or file share that the system running HP SUM can access.	a. Select the location type HP SUM default location. b. Enter the directory path to the baseline, or click Browse and use the menu to navigate to the directory.
UNC path (for example \\host\dir): This location uses UNC paths that the system can access. NOTE: UNC path is only supported in Windows systems.	a. Select the location type UNC path (for example \\host\dir). b. In the Enter URI for the baseline field, type the UNC address for the source baseline.

To add this baseline:	Follow these steps:
	c. Enter the username, required. d. Enter the password, required NOTE: HP SUM does not support mapped UNC drives.
Download from hp.com: This location downloads updates from ftp.hp.com. HP SUM downloads the latest versions of updates that HP has released.	a. Select the location type Download from HP. b. Click Browse and use the menu to navigate to the directory where you want to save the custom baseline. c. Enter the URI for the proxy server. (Only required if you chose to use proxy server.) d. Enter the username. (Only required if you chose to use proxy server.) e. Enter the password. (Only required if you chose to use proxy server.) f. Click Done.

NOTE: HP SUM adds all baselines in a directory.

3. Click **Add**.

Check the activity log to see the status of the baseline.

NOTE: If the baseline does not appear in the baseline list, make sure there are updates in the directory.

NOTE: HP SUM begins to inventory a baseline as soon as you finish adding the baseline. To minimize the impact on system resources, HP SUM does not recommend adding other baselines until the baseline inventory process finishes.

Creating a custom baseline

Creating a custom baseline allows you to deploy specific updates, minimize the size of update baselines or bootable and non-bootable ISOs, and standardize the updates that you deploy to your environment.

NOTE: To create a custom bootable ISO, you must have a bootable HP SPP or HP SUM ISO available in your source baselines.

1. From the Baseline Library screen, click **Actions**→**Create Custom**.
2. In the **Description** text box, enter a description for the baseline. The maximum length for the description is 50 characters.
3. In the **Version** boxes, use the navigation arrows to create a version number, for example 2013.5.21.

NOTE: The Baseline Name is automatically generated by combining the description and version.

4. In the **Output Location** field, click **Browse** to navigate to a directory where you want to save the baseline.

NOTE: HP SUM does not support creating a new directory when it creates the custom baseline. Create the new directory outside of HP SUM.

NOTE: Do not save more than one custom baseline in a directory.

5. Select **Make Bootable ISO file** if you want to create a bootable ISO with your baseline.
If you choose to make a bootable ISO, click **Browse** and choose a directory where HP SUM can find the source for the bootable ISO. The custom baseline bootable ISO includes the version of HP SUM included in the source ISO. For example, if you choose the HP SUM 6 ISO as the source ISO, HP SUM 6 is included in the ISO. If you choose the ISO with the SPP 2013.09.0 as the source ISO, then HP SUM 5.3.5 is included in the ISO.
6. Select whether you want to **Run in background** or not.
If you select to create the baseline in the background, you can monitor the progress in the Activity screen. For more information, see [“Using the Activity screen” \(page 16\)](#).
7. In the **Baseline Sources** section, select the baselines that contain the components you want to include in the custom baseline.

NOTE: If the same component is included in more than one source baseline, HP SUM lists the component multiple times. If you select the component from multiple baselines, HP SUM displays the component multiple times when you add it as a baseline, but the baseline only includes one copy of the component file.

8. In the Filters section, select the **Component Type** you want to choose, **Firmware**, **Software**, or **Both**.
9. In the **Filters** section, select each kind of update you want to include in the baseline:
 - **Critical Updates:** Updates that HP requires you to deploy immediately.
 - **Recommended Updates:** Updates that HP recommends you deploy at your earliest convenience.
 - **Optional Updates:** Update to this version if your system is affected by one of the documented fixes, or if there is a desire to utilize any of the enhanced functionality provided by this version.
10. Select the **Advanced Filters** you want to apply to the baseline:

Filter category	Options
Architecture	Select the system architecture to include in the baseline.
OS Type	Select the operating systems to include in the baseline.
Device Type	Select the device types to include in your baseline.

11. Click **Apply Filters**. HP SUM displays a list of available updates.
12. Select the individual component updates you want to include in your baseline.
Use the search function to find specific updates in the list.
13. Click **Create ISO** or **Save Baseline** to create the baseline.

When you click **Create ISO** or **Save Baseline**, HP SUM does not close the Create Custom Baseline screen, in case you want to create another custom baseline.

NOTE: If you create an ISO, HP SUM also creates a baseline in the directory. You can add this baseline to HP SUM so you can use it locally.

- ① **IMPORTANT:** Do not create more than one custom baseline containing a large number of components at a time.
-

14. Click **Close** to close the Create Custom Baseline screen.

NOTE: HP SUM does not automatically add the custom baseline to the Baseline Library. For more information on adding a baseline, see [“Adding a baseline” \(page 18\)](#).

Deleting a baseline

HP SUM does not support deleting a baseline if it is associated with a node.

To delete a baseline:

1. On the Baseline Library screen, select a baseline.
2. Click **Actions**→**Delete**.

NOTE: If the baseline is associated with one or more nodes, HP SUM displays a list of the nodes. Edit the nodes to change the associated baseline, or delete the node. For more information, see “Editing a node” (page 25) or “Deleting a node” (page 28).

3. Click **Yes, delete** to confirm that you want to delete the baseline.

Configuring components

Some components might have required or optional configuration settings. Configuration parameters include information necessary to set up the component correctly or passwords required for software installed by the component. The configuration is stored within the component and is propagated to all nodes. If the optional configuration data of a component is not provided and the component has not been installed previously, the default values for that configuration data are used. If the component has been previously installed and configured and no changes are made to the configuration data, the existing configuration information is preserved. Component configuration requires that Smart Components are in a write-accessible location. A CD, DVD, or read-only network share is not supported. You can change component options from the Baseline Library screen.

Fixes

Search:

Component	Name	Type	Criticality	Version	Configuration ▼
hpsmb-7.2.2-7.i386.rpm	HP System Management Homepage for Linux (x86)	Software	Recommended	7.2.2-8	Needs configuration
hpsmb-7.2.2-7.x86_64.rpm	HP System Management Homepage for Linux (AMD64/EM64T)	Software	Recommended	7.2.2-8	Needs configuration
cp019016.exe	HP Insight Management WBEM Providers for Windows Server	Driver	Recommended	9.4.0.0	Needs configuration
cp019017.exe	HP Insight Management WBEM Providers for Windows Server x64 Editions	Driver	Recommended	9.4.0.0	Needs configuration
cp020257.exe	HP System Management Homepage for Windows x86	Driver	Recommended	7.2.2.8	Needs configuration

To configure components:

1. On the Baseline Library screen, scroll down to the list of components, and then click **Needs Configuration** for each component you need to configure.

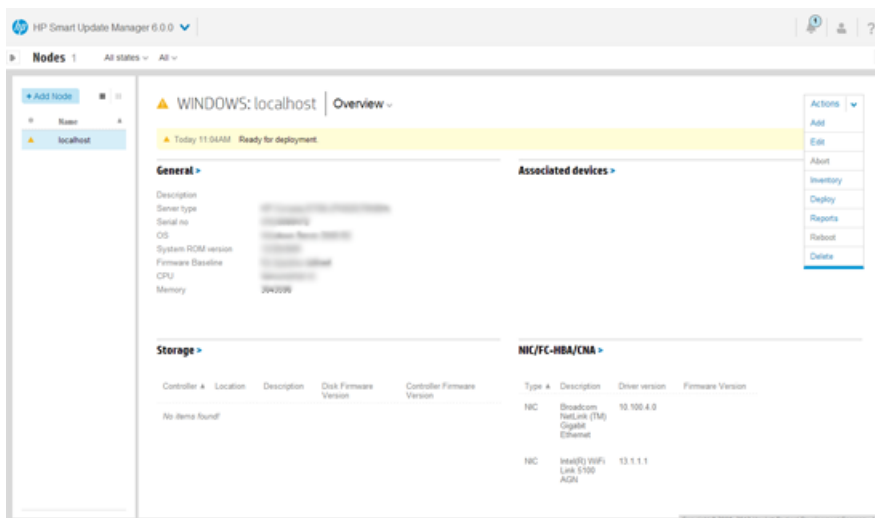
NOTE: Configuration options vary based on the component.

NOTE: HP SUM 6.0 does not support configuring components released prior to the HP SPP 2013.09.0. Use HP SUM 5.3.5 to update these components.

Using the Nodes screen

Nodes are hardware environment components. The Nodes screen displays the nodes that you have added to HP SUM and HP SUM can manage. On the Nodes screen, HP SUM displays four sets of information about the node:

- General
- Associated devices
- Storage
- NIC/FC-HBA/CNA



NOTE: The details that HP SUM displays vary based on the node type.

NOTE: If this is the first time you are running HP SUM 6, and have set up targets in HP SUM 5.x, you can use a migration script to populate the Nodes screen in HP SUM 6.0. For more information, see [“Migrating nodes from HP SUM 5.x to HP SUM 6” \(page 10\)](#).

Selecting multiple nodes

On the Nodes screen, select more than one node by pressing **CTRL** and clicking the other nodes.

If you select multiple nodes, you can:

- View node IP address, node type, and other information
- Generate, set report options, and view reports in the Reports Center
- Choose baselines to use with the nodes
- Set reboot and force options
- Change credentials

Adding a node

You can add a node using a specific IP address, or by searching an IP address range.

Adding a single node by IP address

1. From the Nodes screen, click **Add Node**.
2. Enter the IP address for the new node in the **IP/DNS** field.

The screenshot shows the 'Add New Nodes' interface. It features three main sections: 'Nodes', 'Baseline', and 'Credentials'. The 'Nodes' section contains input fields for 'IP/DNS' and 'Description', a 'Find Nodes' button, a 'Type' dropdown menu currently set to 'Linux', and two radio button options for handling updates. The 'Baseline' section includes searchable fields for 'Baseline' and 'Additional Packages'. The 'Credentials' section offers two radio button options for credential selection and a 'Username' input field. At the bottom right, there are three buttons: 'Add' (highlighted in blue), 'Reset', and 'Cancel'.

3. Enter a description for the node.
4. In the **Type** field, select the node type. If you do not know the node type, select **Unknown**. During the inventory process, HP SUM determines the node type.

NOTE: Selecting the correct node type might help HP SUM complete node inventory faster.

Some nodes allow you to select the check-box so that HP SUM automatically adds Associated Devices. The associated devices identified varies based on the node type. Select what you want HP SUM to do if it discovers updates already running on the node:

- **Leave the update running and bypass the node**
 - **Stop any currently running update process on the node and replace it with updates from this session**
5. If you want, select the baseline, additional package, or both to apply to this node. For more information on adding a baseline, see [“Adding a baseline” \(page 18\)](#).
 6. Select one of the following:
 - **Use current credentials (requires existing trust relationship with the node):** This option is for Windows nodes only.
 - **Enter administrator credentials**
 7. Linux nodes allow you to use sudo credentials to deploy updates without logging into the node with root credentials. To use sudo commands, you have to install sudo capabilities on the node.

If you want to use sudo, in the **Access Level** field, select one of the following.

- Click **Use sudo to update components** if you want to use sudo credentials.
- Click **Enter super user credentials to update components** if you want to enter sudo credentials.

NOTE: Enter the sudo credentials in the Credentials field.

8. Click **Add**.

Adding a node by searching a range of IP addresses

1. From the Nodes screen, click **Add Node**.
2. Click **Find Nodes**.
3. Enter a range of IP addresses for HP SUM to search, and click **Search**.

The screenshot shows the 'NodeRange' dialog box. At the top, there's a title bar 'NodeRange'. Below it, there are two input fields: 'IP Start Address' and 'IP End Address'. A 'Search' button is located below these fields. Underneath the 'Search' button is a section titled 'Available Nodes' which contains a search input field with a magnifying glass icon. Below this is a table header with 'IP Address' and 'DNS Target Name' separated by a dotted line, followed by the text 'No nodes found!'. To the right of this section is an 'Add Nodes' button. Below the 'Add Nodes' button is another table header with 'IP Address' and 'DNS Target Name' separated by a dotted line, followed by the text 'No nodes added!'. At the bottom right of the dialog box is a 'Close' button.

4. Select the nodes you want to add, and then click **Add**.
5. In the **Baselines** section, select a **Baseline**, **Additional Packages**, or both (Optional).
6. In the **Credentials** section, select whether to **Use current user credentials** or **Enter administrator credentials** (Windows only).

If you choose to enter administrator credentials, enter a username and password.

The screenshot shows the 'Credentials' section. It has two radio buttons: 'Use current credentials (requires existing trust relationship with the node)' and 'Enter administrator credentials'. The 'Enter administrator credentials' option is selected. Below these radio buttons are two input fields: 'Username' and 'Password'.

The screenshot shows the 'Access Level' section. It has three radio buttons: 'None', 'Use sudo to update components', and 'Enter super user credentials to update components'. The 'None' option is selected.

7. Click **Add**.
In the **Added Nodes** section, HP SUM displays the nodes you selected.
8. Click **Close** to go back to the Nodes screen.

Editing a node

1. From the Nodes screen, highlight the node and then click **Actions**→**Edit**.
2. Change the items you want to edit:

- **Description**
- **Type**

NOTE: Select the check-box if you want HP SUM to automatically add Associated Devices. (Not available for all node types.)

- HP SUM displays the currently selected baseline and additional package. Use the drop-down menu if you want to change these.

NOTE: If you want to delete a baseline or additional package, the baseline or additional package cannot be associated with a node. Associate a new baseline or delete the node to delete a baseline.

- **Credentials**
- **OS Management Options** – None, Use SNMP, Agentless Management Service, Use WBEM
- **Reboot Options** – Reboot system, Reboot delay, Reboot message
- **Installation Options** – Downgrade or rewrite versions
- **Force Options** – Show firmware or software updates

3. Click **OK** to accept the changes and return to the Nodes screen.

NOTE: Make sure HP SUM has completed all functions, for example, node inventory or deployment, before you edit a node.

Aborting a node update

If you have deployed updates to a node and want to cancel the updates:

1. From the Nodes screen, highlight the node, click **Actions**→**Abort**.

NOTE: If HP SUM has started to perform updates, it completes the component update it is deploying, and then abort the remaining component updates.

Node inventory

The node inventory collects information about the node, and the firmware, software, and driver versions installed on the node.

NOTE: HP SUM cannot inventory a node if the credentials are not valid. If the credentials are not valid for the node, edit the node credentials. For more information, see [“Editing a node” \(page 25\)](#).

To inventory a node:

1. From the Nodes screen, highlight the node and then click **Actions**→**Inventory**.
2. Select a baseline, additional package, or both to apply to the node.
3. Click **Inventory**.

NOTE: If you perform inventory on an OA node, HP SUM automatically adds and inventories the associated nodes if you select the check box on the Add Nodes screen. You need to add VC credentials separately. If you perform inventory on other nodes, HP SUM does not find an associated OA until the end of the inventory process. You need to add credentials to the OA and then perform inventory on the OA node.

Deploying a node

The screenshot shows the 'Deploy' window in HP SUM. It contains the following elements:

- IP/DNS:** localhost
- Type:** WINDOWS
- Installation Options:** A section with expandable options.
- Force options can be applied to:** A section for selecting components to force.
- Associated Node Details:** Includes a search bar and a table with columns: IP Address, Node Type, Component Details, and Status. The table currently shows 'No associated nodes found!'.
- Baseline Library:** Includes a search bar and a table with columns: Package, Ready to proceed, Type, Family, Installed version, Active Version, and Available Version.

Package	Ready to proceed	Type	Family	Installed version	Active Version	Available Version
Headless Server Registry Update for Windows (cp016819)	☒	Driver	Driver			1.0.0.0
HP Insight Management Agents for Windows Server (cp019272)	☒	Driver	Driver			9.40.0.0

At the bottom of the window are three buttons: 'Analysis', 'Deploy' (highlighted in blue), and 'Cancel'.

When you open the Deploy screen, HP SUM displays the IP/DNS address for the node and the node type. To set deployment options:

1. From the Nodes screen, select a node to update, and then click **Actions→Deploy**.

NOTE: The Deploy function only deploys the current partition if you are updating an HP Integrity BL870c i4 and BL890c i4 server. If you want to deploy an enclosure, see [“Deploying all partitions in an HP Integrity BL870c i4 or BL890c i4 node”](#) (page 27).

2. If you want to change installation options, click **Installation Options** and then select from the following:
 - **Downgrade:** This allows you to downgrade all of the components in the node to an older firmware version without clicking **Force** for each component in the baseline library list.
 - **Rewrite:** This allows you to rewrite the current firmware version to components in the node without clicking **Force** for each component in the baseline library list.
3. If you want to downgrade or rewrite options to firmware or software only, click **Force options can be applied to**, and then select from the following:
 - **Firmware:** This displays only firmware updates.
 - **Software:** This displays only software updates.
4. In the **Associated Node Details** field, select any nodes that you want to update.
5. In the **Baseline Library** field, select each component that you want to update. Use the Search box to type in search terms for the components. For more information about an update, click

the version number. If you are deploying a Linux node, HP SUM displays only RPM updates that are valid for your node.

NOTE: You can click **Force** to downgrade or rewrite a component that is available for downgrade or rewrite. If you select **Downgrade**, **Rewrite**, or both in the **Installation Options**, HP SUM does not display the **Force** button.

NOTE: HP SUM supports configuring components on the Baseline Library screen. For more information, see [“Configuring components” \(page 21\)](#).

-
6. If you want to change when the node reboots after an update, click **Reboot Options** and do the following:
 - a. In **Reboot System After Installation**, select **No, If Needed**, or **Always**.
 - b. In **Reboot Delay (seconds)**, enter the number of seconds you want the node to wait before beginning the reboot.
 - c. In the text box, enter a reboot message of up to 255 characters to display before the node reboots.

NOTE: If HP SUM cannot successfully deploy all updates in a node, it does not reboot the node. View the deployment logs to find and resolve the issue before you reboot the node.

-
7. Click **Analyze**, and then click **Deploy** to begin the deployment.

NOTE: To begin the deployment, all dependency issues must be resolved, for example, adding a baseline, administrator credentials, and supported installed versions.

-
8. Click **View log** for the node, in the General section of the Node screen, and then click **View log** for the component you installed, to view the details of the installation.

Deploying all partitions in an HP Integrity BL870c i4 or BL890c i4 node

1. Inventory the HP Integrity BL870c i4 or BL890c i4 node partition.
When HP SUM completes the inventory, it displays associated partitions in the Associated Devices section.

NOTE: You might need to inventory all of the associated nodes before proceeding with deployment.

-
2. Click **Do Deploy Domain** in the yellow status box.
 3. On the Deploy Integrity iLO Domain screen, click **Deploy**.

NOTE: If HP SUM does not display all the associated devices, manually add the nodes. For more information about adding a node, see [“Adding a node” \(page 22\)](#).

-
4. Select the following
 - In the Baselines section, select the baselines you want HP SUM to use.
 - In the Installation Options section, select if you want to downgrade or rewrite the updates.
 - In the Reboot Options, select any partitions you want to reboot after updating.

NOTE: If you make any changes to these sections, click **Analysis**.

-
5. Click **Deploy**.
HP SUM closes the Deploy Integrity iLO Domain screen, and updates the domain.

Node reports

You can generate reports that give details about the node firmware, software, and driver details, components you can install, failed dependencies for nodes, and updates HP SUM installs. The reports for a node enable you to generate HTML, XML, and CSV reports. On the Reports screen, HP SUM includes a **Reports Center** that displays previous reports you have generated.

Reports localhost Select Report Types

Select Report Types

☒ Firmware details

☒ Installable details

☒ Failed dependency details

☐ Installed Details

Select Report Format

☒ HTML

☐ XML

☐ CSV

Advanced Report Options

☐ Run in Background

Reports Center

Generated At	Firmware	Installable	Installed	Failed Dependency	Format	Node(s)	Status
No Reports Available							

Generate Reset Close

To generate a report:

1. From the Nodes screen, highlight a report, and then click **Action→Report**.
2. Select the reports you want to generate:
 - **Firmware details:** This displays firmware versions that are currently installed on the node.
 - **Installable details:** This displays components that HP SUM can install on the node.
 - **Failed Dependency details:** This displays any failed dependencies on the node.
 - **Installed details:** This displays details about what HP SUM installed on a node.

NOTE: If you cannot select a report, HP SUM might not have enough information to generate the report.

3. Click the report format you want HP SUM to generate, **HTML**, **CSV**, or **XML**.
4. Click **Generate**.

Deleting a node

1. From the Nodes screen, highlight the node, and then click **Actions→Delete**.

NOTE: If you want to delete a baseline, the baseline cannot be associated with any nodes. You can delete the node and then delete the baseline. If you delete the node, and want to deploy updates to the node, you will have to re-add the node.

2. Click **Yes, Delete** to confirm you want to delete the node.

Server overview

When you select a node, the Server Overview screen displays a progress bar for inventory and deployment of the node, and information about the node. You can change the baseline for the server on this screen.

During the inventory and deploy process, HP SUM displays a progress bar for current tasks.

NOTE: If you added an Integrity node, see [“Integrity node overview” \(page 29\)](#) for more information.

Integrity node overview

If you added an Integrity node, HP SUM automatically displays the Server Overview screen after HP SUM finishes inventorying and adding the node. HP SUM displays the following information:

- Model
- Associated OA (if the node is a blade)
- Complex firmware version
- Location
- Part number
- Serial number
- List of devices

The following actions are available from the Overview screen:

- **Add:** For more information, see [“Adding an enclosure” \(page 29\)](#).
- **Edit:** For more information, see [“Editing a node” \(page 25\)](#).
- **Report:** For more information, see [“Node reports” \(page 28\)](#).
- **Deploy:** For more information, see [“Deploying a node” \(page 26\)](#).

Using the Enclosures screen

The Enclosures screen displays all of the enclosure nodes that HP SUM has inventoried. You can manage your enclosures from this screen.

Adding an enclosure

1. From the Enclosure screen, click **Add Enclosure**.
2. The Add Enclosure procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing an enclosure

1. From the Enclosures screen, highlight the enclosure you want to edit.
2. Click **Actions**→**Edit**.
3. The Edit Enclosure procedure is the same as the Edit Node function. For more information, see [“Editing a node” \(page 25\)](#).

Aborting an enclosure update

1. From the Enclosures screen, highlight the enclosure whose updates you want to cancel.
2. Click **Actions**→**Abort**.

Enclosure inventory

1. From the Enclosures screen, highlight the enclosure on which you want to take inventory.
2. Click **Actions**→**Inventory**.
3. The Enclosure Inventory procedure is the same as Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying an enclosure

1. From the Enclosures screen, highlight the enclosure you want to deploy.
2. Click **Actions**→**Deploy**.
3. The Enclosure Deploy procedure is the same as the Node Deploy procedure. For more information, see [“Deploying a node” \(page 26\)](#).

Enclosure reports

1. From the Enclosures screen, highlight the enclosure for which you want to generate reports.
2. Click **Actions**→**Reports**.
3. The Enclosure Report procedure is the same as the Node Reports procedure. For more information, see [“Node reports” \(page 28\)](#).

Deleting an enclosure

1. From the Enclosures screen, highlight the enclosure you want to delete.
2. Click **Actions**→**Edit**.
3. Click **Yes, delete** to delete the enclosure.

Using the Servers screen

The Servers screen displays all of the server nodes that HP SUM has inventoried. You can manage your servers from this screen.

Adding a server

1. From the Server screen, click **Add Server**.
2. The Add Server procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing a server

1. From the Server screen, highlight the server you want to edit.
2. Click **Actions**→**Edit**.
3. The Edit Server procedure is the same as the Edit Node procedure. For more information, see [“Editing a node” \(page 25\)](#).

Aborting a server update

1. From the Servers screen, highlight the server whose updates you want to cancel.
2. Click **Actions**→**Abort**.

Server inventory

1. From the Servers screen, highlight the server on which you want to take inventory.
2. Click **Actions**→**Inventory**.
3. The Server Inventory procedure is the same as the Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying a server

1. From the Servers screen, highlight the server you want to deploy.
2. Click **Actions→Deploy**.
3. The Server Deploy procedure is the same as the Node Deploy procedure. For more information, see [“Deploying a node” \(page 26\)](#).

Server reports

1. From the Servers screen, highlight the server for which you want to generate reports.
2. Click **Actions→Reports**.
3. The Server Reports procedure is the same as the Node Reports procedure. For more information, see [“Node reports” \(page 28\)](#).

Deleting a server

1. From the Server screen, highlight the server you want to delete.
2. Click **Actions→Delete**.
3. Click **Yes, delete** to delete the server.

Using the Switches screen

The Switches screen displays only switch nodes that HP SUM has inventoried.

Adding a switch

1. From the Switches screen, click **Actions→Add**.
2. The Add Switch procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing a switch

1. From the Switches screen, highlight the switch, and then click **Actions→Edit**.
2. The Edit Switch procedure is the same as the Edit Node procedure. For more information, see [“Editing a node” \(page 25\)](#).

Aborting a switch update

1. From the Switches screen, highlight the switch whose updates you want to cancel.
2. Click **Actions→Abort**.

Switch inventory

1. From the Switches screen, highlight the switch on which you want to take inventory.
2. Click **Actions→Inventory**.
3. The Switch Inventory procedure is the same as the Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying a switch

1. From the Switches screen, highlight the switch you want to deploy.
2. Click **Actions→Deploy**.
3. The Switch Deploy function uses the same function as the Node Deploy function. For more information, see [“Deploying a node” \(page 26\)](#).

Switch reports

1. From the Switches screen, highlight the switch for which you want to generate reports.
2. Click **Actions→Reports**.
3. The Switch Reports procedure is the same as the Node Reports procedure. For more information, see [“Node reports” \(page 28\)](#).

Deleting a switch

1. From the Switches screen, highlight the switch.
2. Click **Actions→Delete**.
3. Click **Yes, delete**.

Using the VM Hosts screen

The VM hosts screen displays only VM host nodes that HP SUM inventoried.

Adding a VM host

1. From the VM hosts screen, click **Actions→Add**.
2. The Add VM host procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing a VM host

1. From the VM hosts screen, highlight the VM host, and then click **Actions→Edit**.
2. The Edit VM host procedure is the same as the Edit Node procedure. For more information, see [“Editing a node” \(page 25\)](#).

Aborting a VM host update

1. From the VM hosts screen, highlight the VM host whose updates you want to cancel.
2. Click **Actions→Abort**.

VM hosts inventory

1. From the VM hosts screen, highlight the VM host you want to take inventory.
2. Click **Actions→Inventory**.
3. The VM host Inventory procedure is the same as the Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying a VM host

1. From the VM hosts screen, highlight the VM host you want to deploy.
2. Click **Actions→Deploy**.
3. The VM host Deploy procedure is the same as the Node Deploy procedure. For more information, see [“Deploying a node” \(page 26\)](#).

VM host reports

1. From the VM hosts screen, highlight the VM host for which you want to generate reports.
2. Click **Actions→Reports**.
3. The VM host report function uses the same function as the Node report function. For more information, see [“Node reports” \(page 28\)](#).

Deleting a VM host

1. From the VM hosts screen, highlight the VM host.
2. Click **Actions→Delete**.
3. Click, **Yes, delete**.

Using the iLO screen

The iLO screen displays only iLO nodes that you have added to HP SUM.

Adding an iLO

1. From the iLO screen, click **Actions→Add**.
2. The Add iLO procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing an iLO

1. From the iLO screen, highlight the iLO you want to edit.
2. Click **Actions→Edit**.
3. The Edit iLO procedure is the same as the Edit Node procedure. For more information, see [“Editing a node” \(page 25\)](#).

Aborting an iLO update

1. From the iLO screen, highlight the iLO whose updates you want to cancel.
2. Click **Actions→Abort**.

iLO inventory

1. From the iLO screen, highlight the iLO on which you want to take inventory.
2. Click **Actions→Inventory**.
3. The iLO inventory procedure is the same as the Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying an iLO

1. From the iLO screen, highlight the iLO you want to deploy.
2. Click **Actions→Deploy**.
3. The iLO Deploy procedure is the same as the Node Deploy procedure. For more information, see [“Deploying a node” \(page 26\)](#).

iLO reports

1. From the iLO screen, highlight the iLO for which you want to generate reports.
2. Click **Actions→Reports**.
3. The iLO Reports procedure is the same as the Node Reports procedure. For more information, see [“Node reports” \(page 28\)](#).

Deleting an iLO

1. From the iLO screen, highlight the iLO.
2. Click **Actions→Delete**.
3. Click **Yes, delete**.

Using the Intelligent Power Distribution Units screen

The iPDU screen displays only iPDU nodes that you have added to HP SUM.

Adding an iPDU

1. From the iPDU screen, click **Actions→Add**.
2. The Add iPDU procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing an iPDU

1. From the iPDU screen, highlight the iPDU you want to edit.
2. Click **Actions→Edit**.
3. The Edit iPDU procedure is the same as the Edit Node procedure. For more information, see [“Editing a node” \(page 25\)](#).

Aborting an iPDU update

1. From the iPDU screen, highlight the iPDU whose updates you want to cancel.
2. Click **Actions→Abort**.

iPDU inventory

1. From the iPDU screen, highlight the iPDU on which you want to take inventory.
2. Click **Actions→Inventory**.
3. The iPDU Inventory procedure is the same as the Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying an iPDU

1. From the iPDU screen, highlight the iPDU you want to deploy.
2. Click **Actions→Deploy**.
3. The iPDU Deploy procedure is the same as the Node Deploy procedure. For more information, see [“Deploying a node” \(page 26\)](#).

iPDU reports

1. From the iPDU screen, highlight the iPDU for which you want to generate reports.
2. Click **Actions→Reports**.
3. The iPDU Report procedure is the same as the Node Reports procedure. For more information, see [“Node reports” \(page 28\)](#).

Deleting an iPDU

1. From the iPDU screen, highlight the iPDU.
2. Click **Actions→Delete**.
3. Click **Yes, delete**.

Using the Virtual Connects screen

The Virtual Connects screen displays only Virtual Connect nodes that you have added to HP SUM.

Adding a Virtual Connect

1. From the Virtual Connects screen, click **Actions→Add**.
2. The Add Virtual Connect procedure is the same as the Add Node procedure. For more information, see [“Adding a node” \(page 22\)](#).

Editing a Virtual Connect

1. From the Virtual Connects screen, highlight the Virtual Connect you want to edit.
2. Click **Actions**→**Edit**.
3. The Edit Virtual Connect procedure is the same function as the Edit Node procedure. For more information, see [“Editing a node” \(page 25\)](#).

Aborting a Virtual Connect update

1. From the Virtual Connects screen, highlight the Virtual Connect whose updates you want to cancel.
2. Click **Actions**→**Abort**.

Virtual Connect inventory

1. From the Virtual Connects screen, highlight the Virtual Connect on which you want to take inventory.
2. Click **Actions**→**Inventory**.
3. The Virtual Connect Inventory procedure is the same as the Node Inventory procedure. For more information, see [“Node inventory” \(page 25\)](#).

Deploying a Virtual Connect

1. From the Virtual Connects screen, highlight the Virtual Connect you want to deploy.
2. Click **Actions**→**Deploy**.
3. The Virtual Connect Deploy procedure is the same as the Node Deploy procedure. For more information, see [“Deploying a node” \(page 26\)](#).

Virtual Connect reports

1. From the Virtual Connects screen, highlight the Virtual Connect for which you want to generate reports.
2. Click **Actions**→**Reports**.
3. The Virtual Connect Reports procedure is the same as the Node Reports procedure. For more information, see [“Node reports” \(page 28\)](#).

Deleting a Virtual Connect

1. From the Virtual Connects screen, highlight the Virtual Connect.
2. Click **Actions**→**Delete**.
3. Click **Yes, delete**.

4 Using legacy scripts to deploy updates

Command-line interface

You can use the HP SUM command-line interface to script custom installations.

Command-line syntax

The general command-line syntax for HP SUM is as follows:

```
hpsum [/h] [/f]:bundle] [/r[eboot]
```

HP SUM with OA requires a user ID and password to log in. The user ID must be an administrator equivalent ID and not an operator or user equivalent level ID.

On Windows, use a slash before each argument. On Linux, use a hyphen before each argument.

If the `/s[ilent]` argument is not included on the command line, the HP SUM GUI appears.

NOTE: Command-line syntax does not support double-byte character sets. Any messages entered through the command line via a double-byte character set will not be displayed correctly.

Using Linux root credentials

If you run HP SUM from a Linux system where you have not logged in to the system as a root user, you can still update nodes from the CLI/Input file if you use Access level options `sudo` or `super` user.

Prerequisites for using Linux root credentials

- If you have run HP SUM as a root user, remove the temp directory created by HP SUM.
- Make sure you have read/write access permissions to the `/tmp` and `/var` directories.
- If you create a `sudo` user, make sure that you add that user to the `/etc/sudoers` file. The following table shows the privileges and specifications for users.

User	Privilege	Specification
Root	ALL= (ALL)	ALL
Sudo_user	ALL= (ALL)	ALL

- Edit the entry in the `/etc/sudoers` file so the system asks for the `sudo` user password instead of root user password when you run the `sudo` command.

The following table shows the privileges to comment or remove from `/etc/sudoers`.

User	Privilege	Specification
All	ALL= (ALL)	ALL

This often occurs in SUSE Linux systems.

`#Defaults targetpw` # ask for the password of the target user. For example,

⚠ WARNING! Only use this option with `Defaults targetpw`.

- To use super user functionality, configure the user as a super user with all root privileges. You can also use non-root user with a root user to update components.

Switch update commands

G7 and earlier servers do not support AMS agents. The following tables show what updates occur when you use the switch agent commands.

NOTE: To find out if you have any switches installed, use HP SUM to create the Firmware Details Report.

The following table shows what updates occur if no switches are currently installed.

Command	G7 and earlier servers	Gen8 servers
no switches	SNMP	AMS
/no_mgmt	Nothing	Nothing
/use_ams	SNMP	AMS
/use_snmp	SNMP	SNMP
/use_wmi	SNMP and WBEM	WBEM
/use_ams /use_snmp	SNMP	AMS and SNMP
/use_ams /use_wmi	SNMP and WBEM	AMS and WBEM
/use_wmi /use_snmp	WBEM and SNMP	WBEM and SNMP
/use_ams /use_wmi /use_snmp	WBEM and SNMP	AMS, SNMP, and WBEM

NOTE: G7 and earlier servers do not support AMS.

Command-line arguments

HP SUM recognizes the following command-line arguments. If you specify the node and credentials, use the /s[ilent] parameter to proceed with the installation.

You cannot use some arguments, such as /romonly and /softwareonly, together.

Help	Description
/h[elp] or /?	Displays command line help information.

Installation options	Description
/f[orce]	Overrides or downgrades an existing component installation. This produces the same results as /f:software.
/f[orce]:bundle	Overrides or downgrades the existing installation of components in the selected bundle.
/f[orce]:rom	Overrides or downgrades the existing installation of the selected firmware components (applies to firmware only).
/f[orce]:software	Overrides or downgrades the existing installation of the selected software components.
/f[orce]:all	Overrides or downgrades the existing installation of the selected software components, firmware components, and bundles.
/g or /downgrade	Downgrades to an earlier version of firmware for multi-node devices such as hard drives and array controllers (applies to firmware only).

Installation options	Description
/e or /rewrite	Rewrites the current version of firmware for multi-node devices such as hard drives and array controllers (applies to firmware only).
/gui	Starts a GUI session of HP SUM.
/offline_automatic	Runs an HP SUM update automatically in offline mode.
/offline_interactive	Runs an HP SUM update in interactive (GUI) mode.
/s[ilent]	Causes the installation to run silently with no GUI output. All data writes to the log file. Any generated prompts use the default option and continue the installation without user input. If a component requires input before installation (such as configuration information), the component installation fails, and an error message writes to the log file. Failed dependencies are not reported to the user when you are using the /s[ilent] argument. To check for failed dependencies, run HP SUM in GUI mode..
/c[omponent] <component_to_install> or <component1_to_install> <component2_to_install>	Specifies which components to install. Components to install can be specified with or without the /c[omponent] argument. • If you are using the /c[omponent] argument, then only one component can be specified with the argument. However, multiple /c arguments and components can be specified on the same line. • If you do not use the /c[omponent] argument, multiple components can be specified at the same time, but the components must be separated by a blank and listed after all the arguments on the command line. • The components are installed in the order provided unless dependencies between components require installation in a different order. If so, the utility changes the installation order based on the component dependencies to ensure the successful installation of as many components as possible. • Multiple components and bundles can be specified on the same command line. When you are mixing components and bundles on the command line, the filter switches control what components and bundles are installed. • HP SUM only uses the component name. If you type the entire directory path, HP SUM will ignore the path. If you use /use_location and /c, HP SUM checks both the default repository and the directory provided.
/group "group_name"	This argument specifies an already defined group name in the HP SUM GUI.
/b[undle] <bundle_to_install> or <bundle1_to_install> <bundle2_to_install>	This argument specifies bundles to install. Bundles to install can be specified with or without the /b[undle] argument. • If you are using the /b[undle] argument, only one bundle can be specified with the argument. However, multiple /b arguments and bundles can be specified on the same line. • If you do not use the /b[undle] argument, multiple bundles can be specified at the same time, but the bundles must be separated by a blank and listed after all the arguments on the command line.

Installation options	Description
<code>/allow_update_to_bundle</code>	This argument is a filter switch. It enables you to install newer versions of components defined in a PSP, ISP, or firmware bundle. This argument enables these components to replace the older versions of the same component that might have shipped with the bundles.
<code>/allow_non_bundle_components</code>	This argument is a filter switch. It enables you to install components that are not included in the bundle but reside in the directory with the components in the bundle.
<code>/use_latest</code>	This argument is a filter switch for use with bundles. The argument enables the latest version of the bundle to be used when multiple versions of bundles are listed on the command line. If there are no bundles specified on the command line, and multiple bundles are in the directory, the <code>/use_latest</code> argument allows HP SUM to use the bundle that has the latest version for installation.
<code>/use_location "file_share"</code>	Specifies a directory or file share that contains the SPP, HP Integrity Smart Update Firmware Bundles, and components for use with HP SUM. The <i>file_share</i> format must be a mapped file share or UNC format. If this argument is not specified, the directory that contains <code>hpsum.exe</code> or HP SUM is used by default. The logged-in account must have access to this location. The <code>/user</code> and <code>/passwd</code> arguments do not have any effect when you are attempting to access the file share. Use those arguments only when you are connecting to a node system.
<code>/use_web</code>	Selects Download from hp.com on the Add Baseline screen. This enables HP SUM to download updates from ftp.hp.com.
<code>/use_proxy <Proxy server></code>	Enables the inclusion of a proxy server (and port number) to access ftp.hp.com. This parameter must be used with <code>/use_web</code>. For example, <code>/use_web /use_proxy <1.22.33.44:80></code>.
<code>/proxy_script <Proxy script></code>	Enables the inclusion of a proxy script to access ftp.hp.com. This parameter must be used with <code>/use_web</code>. For example, <code>/use_web /proxy_script <autoprox.com></code>.
<code>/proxy_password <password></code>	Sets the password to be used for an authenticating proxy server.
<code>/use_d[ownloaded]</code>	Specifies that the check box for Use last downloaded repository from ftp.hp.com on the Source Selection screen in the GUI is selected. This enables the previously downloaded components to be included in the list of possible updates.
<code>/no_mgmt</code>	Specifies that AMS, SNMP, and WBEM Providers management components are optional on the Components Selection screen. In silent mode, HP SUM will not update any management components.
<code>/use_snmp</code>	Specifies that components that use SNMP protocol and the AMS components are available for installation by default. When the <code>/use_snmp</code> argument is used, the AMS components are required, but the WMI components are optional.
<code>/use_wmi</code>	Specifies that components that use the WMI protocol are available for installation. These components are optional

Installation options	Description
	and are not installed unless this argument is used. When the <code>/use_wmi</code> argument is used, and the <code>/use_snmp</code> & <code>/use_ams</code> arguments are not, the SNMP and AMS components are optional. This argument does not apply to HP Integrity servers.
<code>/use_ams</code>	Specifies which AMS components can be installed along with SNMP components. The AMS and SNMP components are available to select by default. When the <code>/use_ams</code> argument is used, the SNMP components are required, and the WMI components are optional.
<code>/romonly</code>	This filter switch allows you to see only the firmware components needed for installation. When you are using this filter switch, you must exit and restart HP SUM if you want to return to an unfiltered state. Do not use the <code>/romonly</code> argument with the <code>/softwareonly</code> argument. (Applies to firmware only.)
<code>/softwareonly</code>	This filter switch allows you to see only the software components needed for installation. When you are using this filter switch, you must exit and restart HP SUM if you want to return to an unfiltered state. Do not use the <code>/softwareonly</code> argument with the <code>/romonly</code> argument.
<code>/express</code>	HP SUM 6.x does not support <code>/express</code> command. If you scripts contain this command, HP SUM will execute this command the same way it runs <code>/silent</code> .

Overriding errors	Description
<code>/tpmbypass</code> or <code>/ignore_tpm</code>	Specifies that if a TPM is enabled, the warning message should be ignored and component installation can continue. For more information about TPM, see “Disabling BitLocker to permit firmware updates (Windows only)” (page 13).
<code>/ignore_warnings</code>	Allows the installation to proceed on a Superdome 2 node even when warnings are returned during analysis.
<code>/continue_on_error <error></code>	Causes the installation to continue and ignore errors. Valid values are as follows: * <code><error>=ServerNotFound</code> and * <code><error>=BadPassword</code>. * <code><error>=FailedDependencies</code> You can use the <code>ServerNotFound</code> option to bypass inactive or unavailable remote hosts when you are deploying firmware or software to multiple remote hosts at the same time.
<code>/override_existing_connection</code>	Defines the behavior when a remote node has an existing HP SUM session in progress. It overrides the session in progress and re-initializes the installation framework on the remote host.
<code>/On_failed_dependency: <parameter></code>	Provides HP SUM <code><parameter></code> the information on how to proceed when a component has a failed dependency. The supported parameters for this argument are <code>OmitHost</code> (default), <code>OmitComponent</code> , and <code>Force</code> . <code>OmitHost</code> causes the host to be put into a failure state, and no installation is attempted. <code>OmitComponent</code> deselects the affected components and proceeds with any updates that

Overriding errors	Description
	do not have dependency failures. Force attempts all updates, even if they have dependency failures.

Reboot options	Description
<code>/r[eboot]</code>	If the following conditions are met, this argument causes the server (or host server in a remote installation) to reboot: - The <code>/reboot</code> option is selected or given as a command-line argument. - All components selected for installation are successfully installed. - At least one of the installed components requires a reboot to complete its installation.
<code>/reboot_message "reboot message"</code>	Displays the specified reboot message on remote consoles connected to the server that you want to reboot. You must use this argument with the <code>/reboot</code> option, or the argument is ignored.
<code>/reboot_delay timeout_in_secs</code>	Delays the reboot of the server for the length of time that the <code>timeout_in_seconds</code> variable specifies. You must use this argument with the <code>/reboot</code> option, or the argument is ignored. Acceptable values are between 15 and 3600. - The default timeout value is 15 seconds for Microsoft Windows operating systems and 60 seconds for Linux. - For Linux, the reboot delay time is converted from seconds to minutes, and any value under a full minute (59 seconds or less) rounds to the next minute.
<code>/reboot_always</code>	If the following conditions are met, this argument forces the server to reboot: - The <code>/reboot_always</code> option is selected or given as a command-line argument. - All components selected for installation are successfully installed.

Simulating HP SUM	Description
<code>/dryrun</code>	HP SUM 6.0 does not support this command.

Nodes	Description
<code>/user<username> or /username <username></code>	Use this argument to log in to the remote nodes by using the user ID. For the OA node, use the <code>/oa_username</code> argument. The user name needs to belong to the root or administrator group.
<code>/password <password></code> HP SUM also supports <code>passwd</code> and <code>pwd</code>	Use this argument to use this password for the user ID specified in the <code>/user</code> parameter (except for the OA node). The password is used to log in to remote nodes. For the OA node, use the <code>/oa_password</code> argument.
<code>/su_username</code>	Use root (super user) username to start a session, and inventory and update components when the credentials specified in <code>/username</code> and <code>/passwd</code> do not have root privileges to update components.

Nodes	Description
	/su_username cannot be used with /use_sudo access level. NOTE: Specifying /targettype linux reduces inventory time for deployments. This is an optional command line argument.
/su_password	Use root (super user) password to start a session, and inventory and update components when the credentials specified in /username and /passwd do not have root privileges to update components. /su_password cannot be used with /use_sudo access level. NOTE: Specifying /targettype linux reduces inventory time for deployments. This is an optional command line argument.
/use_sudo	Specifies that username and passwd are sudo user credentials. When you specify /use_sudo in the CLI along with username and passwd, then username and passwd are considered as sudo credentials. /use_sudo cannot be used with /su_username and /su_password access level. NOTE: Specifying /targettype linux reduces inventory time for deployments. This is an optional command line argument.
/target "netAddress"	This is the IP address or the DNS name of a remote host, which can be a remote server, remote iLO NIC port, Virtual Connect Ethernet or Fibre Channel Module for c-Class BladeSystem, or BladeSystem OA. When two OAs are in an enclosure, this argument should be the active OA. When specifying the IP address, use either the IPv4 or IPv6 format.
/targettype "type"	Reduces inventory time for scripted deployments. This is an optional command line argument. The following are valid node types: • Server • Windows • Linux • HPUX • EVA or Command View for EVA • FC Switch or Fibre Channel Switch • OA or Onboard Administrator • SUPERDOME 2 or SUPERDOME2 or Superdome 2 Onboard Administrator • iLO • VC or Virtual Connect • HP SAS or HP SAS B/L Interconnect Switch • VMware or VMware Host • iPDU or Intelligent Power Distribution Unit (iPDU) Use /targettype with /target parameter. You can interchange the sequence of /targettype and /target.

Nodes	Description
	If the node name includes spaces, make sure you enclose the name in quotation marks, " ". NOTE: /targettype is an optional command line argument. Use /targettype and /target together.
/current_credential	Enables the credential of the local host to be used as the credential to access the nodes instead of providing the user name and password explicitly for each node. The assumption is that the current credential is valid for the nodes being accessed. (Applies to Windows only)
/oa_username	This argument provides the user name credential for OA associated with VC specified with the target command-line parameter. Only one set of OA credentials can be specified with command-line parameters. You can add multiple VC nodes to command-line parameters with the target parameter only if the credentials of OAs associated with specified VCs are the same. You do not need to provide an OA network address associated with VC. HP SUM queries it from a specified VC node. To update multiple VCs with different user names and passwords, or VCs with OAs that have different credentials, use the corresponding input files OAUDID and OAPWD.
/oa_password	Provides the password credential for OA associated with VC specified with the "node" command-line parameter. Only one set of OA credentials can be specified with command-line parameters. You can add multiple VC nodes to command-line parameters with the "node" parameter only if the credentials of OAs associated with specified VCs are the same. You do not need to provide an OA network address associated with VC. HP SUM queries it from a specified VC node. To update multiple VCs with different user names and passwords, or VCs with OAs that have different credentials, use the corresponding input files OAUDID and OAPWD .

Log files	Description
/logdir "path"	Redirects the output from HP SUM or the HP BladeSystem c-Class OA flash utility to a different directory than the default location. - For Windows components, the default location is %SYSTEMDRIVE%\CPQSYSTEM\hp\log<netAddress> and the redirected location is <path>\hp\log\<netAddress>. - For Linux components, the default location is /var/hp/log/<netAddress> and the redirected location is <path>/hp/log/<netAddress>.
/v[erbose] or /veryv[erbose]	Sets the verbosity level for the HP SUM execution log file, hpsum_execution_log_<date>_<time>.log. It increases the level of detail that is retained in the log file. The default value is normal verbosity.

Generating reports	Description
/report	Generates a report listing of the node summary and describes how the components in the repository affect the node; for example, whether each component applies to the node. The report is generated in HTML and XML with

Generating reports	Description
	file name formats of HPSUM_Report_<date>_<time>.html and HPSUM_Report_<date>_<time>.xml. By default, the files are located in the present working directory where HP SUM is initiated. If that location is write-protected, you can find the files in the same directory as the HP SUM log files.
/inventory_report	Generates a report listing of the components in the specified repository. The report is generated in HTML and XML with file name formats of HPSUM_Inventory_Report_<date>_<time>.html and HPSUM_Inventory_Report_<date>_<time>.xml. By default, the report is located in the present working directory where HP SUM is initiated. If that location is write-protected, you can find the report in the same directory as the HP SUM log files.
/firmware_report	Generates a report listing of the firmware installed and details of the node. The report is generated in HTML and XML with file names of HPSUM_Firmware_Report.html and fwreport.xml in the directory named HPSUM_Firmware_Report_<date>_<time>. By default, the report is located in the present working directory where HP SUM is initiated. If that location is write-protected, you can find the report in the same directory as the HP SUM log files.
/dependency_report	Generates a report listing of the failed dependencies for all nodes.
/installed_report	Generates a report that lists all installed firmware, software, and driver versions installed on all nodes.

NOTE: HP SUM does not install any updates when you generate a report.

Using input files	Description
/inputfile "filename"	Enables you to script the deployment of firmware and software to multiple remote systems at one time. For more information, see "Input files" (page 48) .
/deleteinputfile	Enables you to instruct HP SUM to delete the input file after it has been read.

Superdome 2 server commands	Description
/update_type <type>	Determines which Superdome 2 firmware is updated. You can choose ALL, COMPLEX, or PARTITIONS. The default selection is ALL. Example: /update_type PARTITIONS
/device_list <device[,device]...>	Use this argument when the /update_type argument is PARTITIONS. This argument specifies the subset of partitions or unassigned blades to update when you do not want to update all partitions or unassigned blades. Valid devices are: npar (number) and blade (enc) / (bay) . Do not put any spaces between the parameters.

Superdome 2 server commands	Description
	Example: /device_list npar1,npar2,blade1/1,blade1/2
/reboot_list <npar[,npar]...>	Use this argument when the /update_type is ALL or PARTITIONS. This argument specifies specific partitions to reboot after the partition firmware has been updated. Valid npar values are npar (number) . Do not put any spaces between parameters. Example: /reboot_list npar1,npar2

Command-line examples

The following command-line parameter examples can be executed within these environments:

- Firmware:
 - System ROM
 - Smart Array controller
 - Hard drives
 - iLO
- Software—later version of:
 - HP Insight Diagnostics Online Edition for Windows Server 2003 (cp008097.exe)
 - HP System Management Homepage (HP SMH) for Windows (cp008257.exe)
- HP SUM
 - Defined groups: Management Servers—three servers (Management Server1, Management Server2, Management Server3)

Example 1:

This command-line input deploys the latest PSP and firmware components:

```
hpsum /use_latest /allow_non_bundle_components /silent
```

Results: All software components and firmware components from the update bundle, which HP SUM determined must be installed, were installed.

Example 2:

Either of the following command-line inputs can deploy the previous version of the PSP only and force all the components to be installed:

- hpsum /f:bundle /softwareonly BP000315.xml
- hpsum /b BP000315.xml /f:bundle /softwareonly

Results: All software components and firmware components from the update bundle, which HP SUM determined must be installed, were installed. No firmware was installed.

Example 3:

This command-line input deploys firmware:

```
hpsum /romonly
```

Results: All firmware components that must be installed, were installed. No software was installed.

Example 4:

Either of the following command-line inputs can deploy two software components:

- `hpsum /f:software cp008097.exe cp008257.exe`
- `hpsum /c cp008097.exe /c cp008257.exe /f:software`

Results: The two components were installed. No firmware or other software was installed.

Example 5:

Either of the following command-line inputs can deploy the latest PSP, later versions of components in the bundle, and firmware to three remote hosts and force all components to be installed:

- `hpsum /group "Management Servers" /current_credential /use_latest /allow_update_to_bundle /allow_non_bundle_components /force:all /override_existing_connection /continue_on_error ServerNotFound /silent /logdir "Management_Server_Files"`
- `hpsum /target "Management Server1" /target "Management Server2" /target "Management Server3" /user administrator /passwd letmein /use_latest /allow_update_to_bundle /allow_non_bundle_components /force:all /override_existing_connection /continue_on_error ServerNotFound /silent /logdir "Management_Server_Files"`

Example 6:

To use sudo to update components:

```
/hpsum / target 10.0.1.7 /targettype linux /username userid /passwd password /use_sudo /silent
```

To use super user to update components:

```
/hpsum /target 10.0.1.7 /targettype linux /username userid /passwd password /silent /su_username root /su_password rootpwd
```

Results: All software components and firmware components from the update bundle, cp008097.exe, and cp008257.exe were installed on Management Server1, Management Server2, and Management Server3.

Return codes

HP SUM has consolidated return codes from Linux and Windows smart components into a new, enhanced return-code mapping. These return codes determine the status of the component installation. You can also use return codes in a script to control the execution of the script and determine any required branching.

In Linux, the negative return codes are reported. These return codes are determined by subtracting the negative value from 256.

Return code	Value	Linux	Windows	Text
SUCCESS_NO_REBOOT	0	0	0	The installation was successful.
SUCCESS_REBOOT	1	1	1	The installation was successful, but a reboot is required.
SUCCESS_NOT_REQUIRED	3	3	3	The component was current or not required.
FAILURE_GENERAL	-1	255	-1	A general failure occurred. For details, see the error log.
FAILURE_BAD_PARM	-2	254	-2	A bad input parameter was encountered.
FAILURE_COMPONENT_FAILED	-3	253	-3	The installation of the component failed.

Windows smart-component return codes

Return code	Meaning
0	The smart component was not installed. For more information, see the log file.
1	The smart component was installed successfully.
2	The smart component was installed successfully, but the system must be restarted.
3	The installation was not attempted because the required hardware is not present, the software is current, or there is nothing to install.

Linux smart-component return codes

Single-node servers:

Return code	Meaning
0	The smart component was installed successfully.
1	The smart component was installed successfully, but the system must be restarted.
2	The installation was not attempted because the required hardware is not present, the software is current, or there is nothing to install.
3	The smart component was not installed. For more information, see the log file.

Multi-node servers:

Return code	Meaning
0	The installation of the deliverable was successful. No reboot is required.
1	The installation of the deliverable was successful. Reboot is required for the deliverable to be enabled.
2	The installation was not attempted because the version to be installed matches the version already installed.
3	The installation was not attempted because of one of the following: • The version to be installed is older than the version already installed. • The supported hardware is not present, not enabled, or in a state that an installation could not be attempted. • The smart component does not support the environment. • There is nothing for the component to accomplish.
4	If the component is being installed to a remote node, such as an OA or other network-based deployment, this return code indicates that the node cannot be found.
5	A user canceled the installation before anything could be installed.

Return code	Meaning
6	The installer cannot run because of an un-met dependency or installation tool failure.
7	The actual installation operation (not the installation tool) failed.

Linux RPM return codes

Return code	Meaning
0	The Linux RPM installation was successful.
1	The Linux RPM installation failed.

VMware ESXi smart-component return codes

Return code	Meaning
0	The installation of the deliverable was successful. No reboot is required.
1	The installation of the deliverable was successful. Reboot is required for the deliverable to be enabled.
2	The installation was not attempted because the version to be installed matches the version already installed.
3	The installation was not attempted because of one of the following: • The version to be installed is older than the version already installed. • The supported hardware is not present, not enabled, or in a state that an installation could not be attempted. • The smart component does not support the environment. • There is nothing for the component to accomplish.
4	If the component is being installed to a remote node, such as an OA or other network-based deployment, this return code indicates that the node cannot be found.
5	A user canceled the installation before anything could be installed.
6	The installer cannot run because of an un-met dependency or installation tool failure.
7	The actual installation operation (not the installation tool) failed.

Input files

HP SUM provides the ability to script the update of multiple, individual nodes or groups of nodes (HP ProLiant and Integrity servers and options) within a single operation through the input file functionality. To protect your credentials, use a secure server or a management console.

To create an input file, use a text editor. All section headers and trailers must match. Failure to use the option causes the GUI mode to be used, but the information provided enables you to skip screens where information has already been provided. You can use the `DRYRUN=YES` option to perform dry runs of installations to ensure the scripts are working without deploying the firmware updates that might be required on each node. Remove the `DRYRUN=YES` option to perform the updates.

For parameters that can take list values, list separators can be commas, semicolons, or spaces.

NOTE: The credentials can be omitted from the file for greater security and passed on the command line to HP SUM. The only limitation that the user ID and credentials must be the same on all.

When the file has been created, to use it with HP SUM, add it as the `inputfile <filename>` parameter to a HP SUM command line. For example, if the name of the input file is `hpsum.in`, the command-line syntax is `hpsum -inputfile hpsum.in`. Full paths can be added to the input file location if the file is not stored in the same location as the HP SUM executable files. The `<filename>` field can be enclosed in double quotes to enable paths with spaces. Also, the input file itself might contain the same options on the command line. The usual command-line options can still be used with the `-inputfile` option and they take precedence over any given input file.

Input file format and rules

The input file is divided into two sections:

- Configuration

The configuration section starts from the beginning of the file and proceeds until the first node section is encountered. This section consists of a number of settings and their values. Each configuration setting must appear on a fresh line in the file, along with its value. Comments start with a pound (#) character at the beginning of the line. Only one pound character is allowed on any line.

- Target

You can provide remote host nodes to HP SUM. This section can repeat any number of times in the input file, providing a way to organize nodes in related sets.

The section starts with the following special header enclosed in brackets:

```
[TARGETS]
```

The section ends with the following special string enclosed brackets:

```
[END]
```

The keyword `TARGETS` can be suffixed with an optional arbitrary string. This string enables you to tag the purpose of the `TARGETS` section. Other than the visible difference in the header, the contents of such a section are not treated any differently. For example:

```
[TARGETS_WIN2003]
```

```
...
```

```
[END]
```

- Credentials: The `TARGETS` section allows the nodes to be grouped according to the credentials needed for logging in remotely. Each `TARGETS` section must have a set of login credentials, which applies to all nodes in that section. If you want to use the current host's login credentials to log in to one or more remote nodes, you can do so by setting the variable `USECURRENTCREDENTIAL` to `YES`. You can supply login credentials for one or more hosts by using the variables `UID` and `PWD`. If you want to provide the variables at the beginning of a `TARGETS` section, use both of them. If you want to provide the variables in the middle of a `TARGETS` section, use one or the other to override the selected variable and continue using the active value for the remaining variable.
- Remote node: You can specify a remote node by using the variable `HOST`. Possible values are a DNS name or an IP address.

File encoding

To allow for the inclusion of double-byte characters, the input file is in UTF-8 format.

Error reporting

If errors occur in the input file, HP SUM exits with a return value of -2 (bad parameter). The details of the location and nature of the error are recorded in `hpsum_execution_log_<date>_<time>.raw`.

Input file parameters

Parameter	Description	Possible values
SILENT	This parameter causes the installation to run silently without GUI or console output. All data is written to the log file. Any generated prompts use the default option and continue the installation without user input. If a component requires input before installation (such as configuration information), the component installation fails (unless the <code>IGNOREERRORS = "FailedDependencies"</code> parameter is supplied), and an error message is written to the log file.	YES, NO
TARGETTYPE	This parameter specifies the node type and can shorten the inventory process.	<code>/targettype vc</code> The following are valid node types: • Server • Windows • Linux • HPUX • EVA or Command View for EVA • FC Switch or Fibre Channel Switch • OA or Onboard Administrator • SUPERDOME 2 or SUPERDOME2 or Superdome 2 Onboard Administrator • iLO • VC or Virtual Connect • HP SAS or HP SAS B/L Interconnect Switch • VMware or VMware Host • iPDU or Intelligent Power Distribution Unit (iPDU) If you use the command with a group, the HP SUM assumes all nodes in the group are the same node type.
FORCEALL	Forces updates to both firmware and software components.	YES, NO
FORCEROM	Forces updates to firmware components.	YES, NO

Parameter	Description	Possible values
FORCESOFTWARE	Forces updates to software components.	YES, NO
FORCEBUNDLE	Enables you to override or downgrade an existing installation of components in the selected bundle.	YES, NO
DOWNGRADE	Downgrade to an earlier version of firmware for multi-node devices such as hard drives and array controllers. (Applies to firmware only.)	YES, NO
REWRITE	Rewrite the same version of firmware only for multi-node devices such as hard drives and array controllers. (Applies to firmware only.)	YES, NO
REBOOTALLOWED	Enables you to reboot, if required.	YES, NO
REBOOTMESSAGE	Create a message to be displayed prior to rebooting.	Any string (not exceeding 256 characters)
REBOOTDELAY	Delay before rebooting.	Time in seconds
COMPONENTSLIST	Limit the list of components to be updated.	Component names with file extensions (.exe, .rpm, or .scexe)
BUNDLESLIST	Limit the list of bundle xml files to be filtered.	Bundle file names
ALLOWUPDATEBUNDLE	This is a filter switch and enables you to install newer versions of components defined in a PSP, ISP, or bundle. This parameter enables these components to replace the older versions of the same component that might have shipped with the bundles.	YES, NO
SKIPTARGET	Defines the behavior when a remote node has an existing HP SUM session in progress. Use this parameter to skip the host if an existing HP SUM session already exists. A value of NO overrides the session in progress and reinitializes the installation framework on the remote host. NOTE: If an HP SUM session discovers a remote node is running HP SUM locally, HP SUM ignores the SKIPTARGET command and skips the remote node.	YES, NO
IGNOREERRORS	Causes the installation to continue and ignore errors. The ServerNotFound option can be used to bypass inactive or unavailable remote hosts when deploying firmware or software to multiple remote hosts at the same time. Use FailedDependencies to ignore any failed dependencies and proceed with the ones that are ready to be installed.	ServerNotFound, BadPassword, FailedDependencies

Parameter	Description	Possible values
SOURCEPATH	Provide a single local baseline path. This action creates an inventory from the given path instead of the local or default baseline.	Directory path
USELATEST	This is a filter switch for use with bundles. The parameter enables you to use the latest version of the bundle when multiple versions of bundles are listed on the command line. If no bundles are specified on the command line, and multiple bundles are in the directory, this parameter enables HP SUM to use the bundle that has the latest version for installation.	YES, NO
DRYRUN	Simulates the installation for a test run. Nothing is installed.	YES, NO
OPTIONS	Specify the HP SUM CLI options inside the input file, which overrides the configuration settings. Parameters can be separated by a semi-colon, comma, or a space. This parameter replaces the <code>LSOPTIONS</code> parameter that was previously supported with LDU.	One or more CLI switch
NOMGMT	Specifies that components that use AMS and WBEM Providers listed as optional updates on the Components Selection screen. In silent mode, HP SUM does not update AMS or WBEM Providers. NOTE: You can only configure SNMP in HP SUM GUI mode.	Yes No
USEWMI	Specifies that components that use WMI protocol are available to be selected for installation. These components are optional by default and are not installed unless this parameter is used. When the <code>/usewmi</code> parameter is used. This parameter does not apply to HP Integrity servers.	YES, NO
USEAMS	Specifies that AMS agentless management service components are available to be selected for installation. This option will only apply to Gen8 and later servers. If this parameter is set for a ProLiant G7 and earlier server model, it will be ignored. These components are only installed by default on Gen8 and later servers. This parameter does not apply to HP Integrity servers.	YES, NO

Parameter	Description	Possible values
ROMONLY	This is a filter switch that allows you to view only the firmware components required for installation. Do not use the /romonly parameter with the /softwareonly parameter.	YES, NO
SOFTWAREONLY	This is a filter switch that allows you to view only the software components required for installation. Do not use the /softwareonly parameter with the /romonly parameter.	YES, NO
USECURRENTCREDENTIAL	Enables the credentials of the local host to be used as the credentials to access the nodes instead of providing the user name and password explicitly for each node. The assumption is that the current credentials are valid for the nodes that are being accessed. (Applies to Windows only.)	YES, NO
WEBUPDATENEEEDED	Instructs HP SUM to include the components from the HP FTP site (ftp://ftp.hp.com) in the list of possible updates.	YES, NO
USEPROXYSERVER	Use a proxy server (and port number) to access the HP FTP site (ftp://ftp.hp.com).	String value For example, 11.22.33.44:80
USEPROXYSCRIPT	Include a proxy script to access the HP FTP site (ftp://ftp.hp.com).	Web URL (for example, autoproxy.com)
DELETEINPUTFILE	Instructs HP SUM to delete the input file after it has been read.	YES, NO (default)
ONFAILEDDEPENDENCY	This parameter instructs HP SUM how to proceed when a component has a failed dependency. The default of OmitHost causes the host to be put in a failure state, and no installation is attempted on it. OmitComponent clears the affected components and proceeds with any updates that do not have dependency failures. Force attempts all updates, even if they have dependency failures.	OmitHost (default), OmitComponent, Force
HOST	This parameter is the IP address or the DNS name of a remote server, remote iLO NIC port, Virtual Connect Ethernet or Fibre Channel Module for c-Class BladeSystem, or BladeSystem OA. When two OAs are in an enclosure, this parameter is the active OA. When specifying the IP address, you can use either the IPv4 or IPv6 format. This parameter specifies an already defined group name in the HP SUM GUI.	IP address, DNS name

Parameter	Description	Possible values
NODE	This parameter is the IP address or the DNS name of a remote server, remote iLO NIC port, Virtual Connect Ethernet or Fibre Channel Module for c-Class BladeSystem, or BladeSystem OA. When two OAs are in an enclosure, this parameter is the active OA. When specifying the IP address, you can use either the IPv4 or IPv6 format. This parameter specifies an already defined group name in the HP SUM GUI.	IP address, DNS name
UID	Logs in to the nodes with your user ID.	<username>
PWD	Uses the password for the user ID specified in the UID. The password is used to log in to nodes.	<password>
LOGFILENAME = "path"	Redirects the output from HP SUM or the HP BladeSystem c-Class OA flash utility to a different directory than the default location. * For Windows components, the default location is %SYSTEMDRIVE%\CPQSYSTEM\hp\log<netAddress> and the redirected location is <path>\hp\log\<netAddress>. * For Linux components, the default location is /var/hp/log/<netAddress> and the redirected location is <path>/hp/log/<netAddress>.	Log file name
CMAMGMTSTATIONRWIPORDNS	Specifies the IP address or DNS host name of a system with read/write access to serve as a management station. You can specify multiple locations separated by a space. (Applies to Linux SPP only.)	IP address, DNS name
CMAMGMTSTATIONROIPORDNS	Specifies the IP address or DNS host name of a system with read-only access to serve as a management station. You can specify multiple locations separated by a space. (Applies to Linux SPP only.)	IP address, DNS name
CMASYSCONTACT	Specifies a person or phone number for administration of this system. (Applies to Linux SPP only.)	String value
CMASYSLOCATION	Designates the location of this system. (Applies to Linux SPP only.)	String value
CMASRTWEBAGENT	Determines whether the HP Systems Insight Manager Web Agent is started when the health application loads. (Applies to Linux SPP only.)	• YES (start the web agent) • NO (do not start the web agent)
CMASRTSTORAGEAGENT	Determines whether the HP Systems Insight Manager Storage Agent is	• YES (start the storage agent) • NO (do not start the storage agent)

Parameter	Description	Possible values
	started when the health application loads. (Applies to Linux SPP only.)	
CMASRTNICAGENT	Determines whether the HP Systems Insight Manager NIC agent is started. (Applies to Linux SPP only.)	- YES (start the web agent) - NO (do not start the web agent)
CMANOTAINTEDKERNEL	Determines whether the HP Lights-Out management driver is started when the health application loads. (Applies to Linux SPP only.)	- YES (start the HP Lights-Out management driver) - NO (do not start the HP Lights-Out management driver)
HPVCAVCRMSEVER	Informs the VCA of the name of the VCRM to use as a software distribution baseline. (Applies to Linux SPP only.)	VCRM name
FORCE-OVERWRITE	HP Systems Management Homepage (hpsmh) uses this parameter to force overwrite the SMH settings of an existing configuration file.	YES, NO (default)
ADMIN-GROUP	HP Systems Management Homepage (hpsmh) uses this parameter to set up security for the web server. (Applies to Linux SPP only.)	Up to five Linux groups, separated by spaces or semicolons, to enable administrative access to the web services.
USER-GROUP	HP Systems Management Homepage uses this parameter to set up security for the web server. (Applies to Linux SPP only.)	Up to five Linux groups, separated by spaces or semicolons, to enable user-level access to the web servers.
OPERATOR-GROUP	HP Systems Management Homepage uses this parameter to set up security for the web server. (Applies to Linux SPP only.)	Up to five Linux groups, separated by spaces or semicolons, to enable operator-level access to the web servers.
ANONYMOUS-ACCESS	Determines whether an anonymous user can access HP Systems Management Homepage. (Applies to Linux SPP only.)	YES, NO (default)
IP-BINDING	HP Systems Management Homepage uses this parameter to determine whether HP SMH can use all available NICs and detect subnets for its web services. (Applies to Linux SPP only.)	YES, NO (default)
IP-BINDING-LIST	HP Systems Management Homepage uses this parameter to restrict the NICs and subnets to use for its web servers. The IP-BINDING parameter must be set to yes for this parameter to be used during installation. (Applies to Linux SPP only.)	IP address pairs separated by semicolons (for example, 10.1.1.1/255.255.255.0; 10.2.2.2/255.255.255.0)
IP-RESTRICTED-LOGINS	HP Systems Management Homepage uses this parameter to restrict login access. (Applies to Linux SPP only.)	YES, NO (default) To enable restrictions on who can log in to the web server, this parameter must be set to yes, and values must be provided to the IP-RESTRICTED-EXCLUDE or IP-RESTRICTED-INCLUDE parameters.

Parameter	Description	Possible values
IP-RESTRICTED-EXCLUDE	HP Systems Management Homepage uses this parameter to exclude specific IP address/netmask pairs from logging into the web services. (Applies to Linux PSP only.) This parameter is ignored unless the IP-RESTRICTED-LOGINS parameter is set to yes.	List of IP address ranges separated by semicolons (for example, 10.1.1.1-10.1.1.10; 10.2.2.2-10.2.2.10)
IP-RESTRICTED-INCLUDE	HP Systems Management Homepage uses this parameter to enable login only from the IP address/netmask pairs specified. (Applies to Linux SPP only.) This parameter is ignored unless the IP-RESTRICTED-LOGINS parameter is set to yes.	List of IP address ranges separated by semicolons (for example, 10.1.1.1-10.1.1.10; 10.2.2.2-10.2.2.10)
LOCALACCESS-ENABLED	HP Systems Management Homepage uses this parameter to determine whether to enable local anonymous access to the web services. (Applies to Linux SPP only.)	• YES (default to include anonymous access) • NO
LOCALACCESS-TYPE	HP Systems Management Homepage uses this parameter to determine the type of access granted to local users. (Applies to Linux SPP only.)	• Anonymous (default) • Administrator CAUTION: Selecting local access with administrator privileges as the login provides full access to any user who has access to the local console, without prompting for a user name or password.
TRUSTMODE	HP Systems Management Homepage uses this parameter to set up the trust relationship mode. (Applies to Linux SPP only.)	• TrustByCert—If you use this value, you must define the CERTLIST parameter to enable access to the server. • TrustByName—If you use this value, you must define the XENAMELIST. • TrustByAll—HP does not recommend using this value because of possible negative security consequences. CAUTION: The accepted values are case-sensitive and must be capitalized as shown. Failure to do so prevents the trust relationship from being set up properly during installation and might affect access to the web server.
CERTLIST	Enables you to provide a list of certificate files or servers where certificates can be obtained for trust relationships for the HP Systems Management Homepage. (Applies to Linux SPP only.)	Certificate file name or Server DNS name

Parameter	Description	Possible values
XENAMELIST	Enables you to provide a list of servers, separated by semicolons, for trust relationships for the HP Systems Management Homepage. (Applies to Linux SPP only.) This parameter is valid only if the TRUSTMODE parameter is set to TrustByName. (Applies to Linux SPP only.)	Server DNS name
HPQLA2X00FO	The hp_qla2x00 QLogic Fibre Channel Driver uses this parameter to determine the failover mode to use. (Applies to Linux SPP only.)	- SinglePath - SecurePath - QLogicFailure NOTE: No default value
HPQLA2X00FORCE	The hp_qla2x00 QLogic Fibre Channel Driver uses this parameter to determine whether to skip detection of third-party storage. (Applies to Linux SPP only.)	Y, N (default)
OAUID	Provides the user name credentials for the OA associated with VC. You must define a value of these variables before the HOST variable in the [TARGETS] section. This parameter applies only for VC firmware. You can also use OAUSER or OAUSERNAME.	User can define the OAUID variable multiple times before each HOST variable.
OAPWD	Provides the password credentials for the OA associated with VC. You must define a value of these variables before the HOST variable in the [TARGETS] section. This parameter applies only for VC firmware. You can also use OAPASSWORD.	User can define OAPWD variable multiple times before each HOST variable.
IGNOREWARNINGS	Allows the installation to proceed on a Superdome 2 node even when warnings are returned during analysis.	YES, NO Superdome 2 servers only
UPDATETYPE	This argument determines which Superdome 2 firmware is updated.	ALL, COMPLEX, PARTITIONS Superdome 2 servers only
DEVICELIST	Use this argument when the UPDATETYPE=PARTITIONS. This argument specifies the subset of partitions or blades to update when you do not want to update all partitions. Valid devices are: npar (number) and blade (enc) / (bay) .	For example: DEVICELIST=npar1,blade1/1 Do not put spaces between the parameters Superdome 2 servers only
REBOOTLIST	Use this argument when the UPDATETYPE= ALL or UPDATETYPE= PARTITIONS. This argument specifies specific partitions to reboot after the partition firmware has been updated. Valid npar values are npar (number) .	For example: REBOOTLIST=npar1,npar2 Do not put spaces between the parameters Superdome 2 servers only

Parameter	Description	Possible values
SUUSERNAME	Use this argument to provide a superuser username. SUUSERNAME cannot be used with USESUDO access level. NOTE: Specifying TARGETTYPE = linux reduces inventory time for deployments. This is an optional input file argument.	<superusername>
SUPASSWORD	Use this argument to provide a superuser password. SUPASSWORD cannot be used with USESUDO access level. NOTE: Specifying TARGETTYPE = linux reduces inventory time for deployments. This is an optional input file argument.	<superuserpassword>
USESUDO	Use this argument to use the sudo command. When you specify USESUDO = YES in the input file, along with UID and PWD, UID and PWD are considered sudo users. USESUDO cannot be used with SUUSERNAME and SUPASSWORD access level. NOTE: Specifying TARGETTYPE = linux reduces inventory time for deployments. This is an optional input file argument.	Yes, No

Examples of the HP SUM input file include the following:

```

REBOOTALLOWED = YES
REBOOTREQUIRED = NO
REBOOTMESSAGE = "Server is going down for a reboot"
REBOOTDELAY = 15
COMPONENTSLIST = cp001234.exe, cp001235.exe
BUNDLESList = bp001234.xml
ALLOWUPDATEBUNDLE = YES
SKIPTARGET = NO
IGNOREERRORS = ServerNotFound, FailedDependencies
SOURCEPATH = c:\pkgsource1
USELATEST = YES
SILENT = YES
OPTIONS = /f:rom
[TARGETS]
HOST = schinta1
HOST = schinta2
UID = root

```

```
PWD = root123
HOST = 234.567.765.432
[END]
USAGE: hpsum /inputfile <path:\inputfile.txt>
```

Examples of inputfile.txt file:

Example: The two nodes are passed to be updated. The nodes do not necessarily have to be OAs. They can be any node that HP SUM supports.

```
DRYRUN = YES
SILENT = YES
[TARGETS]
HOST = BL465C-01
HOST = 192.168.1.2
[END]
```

Example A host DNS is passed along with the user ID and password to use for the hosts in the group.

```
DRYRUN = YES
SILENT = YES
[TARGETS]
HOST = BL685cG6
UID = Bigboss2
PWD = password
[END]
```

Example

```
SILENT = YES
IGNOREERRORS = ServerNotFound,BadPassword, FailedDepedencies
SKIPTARGET = NO
SOURCEPATH = C:\fwcd\firmware-8.70-0\hp\swpackages
[GROUPS]
HOST=winserver
UID=Userid
PWD=password
[END]
```

Example

```
SILENT = YES
IGNOREERRORS = ServerNotFound,BadPassword, FailedDepedencies
SKIPTARGET = NO
SOURCEPATH = C:\ fwcd\firmware-8.70-0\hp\swpackages
FORCEALL = YES
REBOOTALLOWED = YES
REBOOTDELAY = 30
REBOOTMESSAGE = "Install complete, server will reboot in 30 seconds"
[TARGETS]
```

```
HOST=16.83.62.141
```

```
UID=Userid
```

```
PWD=password
```

```
[END]
```

```
[TARGETS]
```

```
HOST=16.83.61.48
```

```
UID=Userid
```

```
PWD=password
```

```
[END]
```

```
[TARGETS]
```

```
HOST=16.83.62.196
```

```
UID=Userid
```

```
PWD=password
```

```
[END]
```

```
[TARGETS]
```

```
HOST=16.83.61.24
```

```
UID=Userid
```

```
PWD=password
```

```
[END]
```

Example: Superdome 2 input files:

```
IGNOREWARNINGS = TRUE
```

```
[TARGET] HOST = 10.0.0.206
```

```
UID = Userid
```

```
PWD = password
```

```
UPDATETYPE = PARTITIONS
```

```
DEVICELIST=npa1,npa2,blade1/4,blade 1/5
```

```
REBOOTLIST=npa1,npa2
```

```
[END]
```

Usage example of access level.

USAGE: hpsum /inputfile <path:\inputfile.txt>

Example: Using sudo to update components on Linux:

```
SILENT = YES
```

```
[TARGETS]
```

```
HOST = 10.0.1.7
```

```
UID = Userid
```

```
PWD = password
```

```
[END]
```

Example: To update multiple targets using sudo in an input file:

```
SILENT = YES
```

```
USESUDO = YES
```

```
[TARGETS]
```

```
HOST = 10.0.1.7
UID = Userid
PWD = password
[END]
[TARGETS]
HOST = 10.0.1.73
UID = Userid
PWD = password
[END]
```

Example: Using super user to update Linux components:

```
SILENT = YES
[TARGETS]
HOST = 10.0.1.7
UID = Userid
PWD = password
SUUSERNAME = root
SUPASSWORD = rootpwd
[END]
```

Example: To update multiple targets using super user in an input file when all targets have the same super user credentials:

```
SILENT = YES
SUUSERNAME = root
SUPASSWORD = rootpwd
[TARGETS]
HOST = 10.0.1.7
UID = Userid
PWD = password
[END]
[TARGETS]
HOST = 10.0.1.73
UID = Userid
PWD = password
[END]
```

Reports

HP SUM generates three types of reports about the specified system or repository. HP SUM generates these reports by using the command-line arguments. If no additional arguments are specified on the command line, HP SUM uses the local host and default repository locations (the directory where HP SUM was initiated). You can specify a node if you provide the appropriate credentials. You

can specify a repository by using other command-line parameters to generate reports. For specific commands, see “[Command-line interface](#)” (page 36).

Report type	Description	Report file information*
/report	This argument generates a report listing, a node summary, and description of how the components in the repository affect the node. For example, whether each component applies to the node. Usage: hpsum /report	The following report files are generated: • hpsum_Report_<date>.html • hpsum_Report_<date>.xml • hpsum_Report_<date>.csv
/inventory_report	This argument generates a report listing of the components in the specified repository. Usage: hpsum /inventory_report	The following report files are generated: • hpsum_Inventory_Report_<date>.html • hpsum_Inventory_Report_<date>.xml • hpsum_Inventory_Report_<date>.csv
/firmware_report	This argument generates a report listing of the firmware in the specified repository. Usage: hpsum /firmware_report	The following report files are generated: • HPSUM_Firmware_Report_<date> • fwreport.xml is placed in a folder named HPSUM_Firmware_Report_<date>
/dependency_report	This argument generates a report that lists failed dependencies for all nodes.	The following report file is generated: HPSUM_FailedDependency_Reports_<datetime>.xml, .html, or .csv.
/installed_report	This argument generates a report that lists all installed firmware, software, and driver versions installed on nodes.	The following report file is generated: HPSUM_Installed_Report_<datetime>.xml , .html, or .csv.

HP SUM generates the reports as XML or HTML files that you can view in a JavaScript-enabled web browser. The supported browsers for viewing the report files are Microsoft Internet Explorer 6.0 and Mozilla Firefox 3.5 and later. By default, the reports are located in the current working directory from where HP SUM was initiated. If that current location is write-protected, the reports are located in the same directory as the HP SUM log files.

HP SUM generates reports in CSV format that you can open in any application that supports CSV format.

The HP SUM log files are located in these folders:

- On Windows operating systems: C:\cpqsystem\hp\log
- On Linux: /var/hp/log

When HP SUM generates the report, the HP SUM GUI does not appear. When the report is generated, HP SUM displays the report file location.

Downloading HP SUM and components from the SDR

If you are using a Linux system, you can download HP SUM as an RPM from the HP SDR. This allows you to download and install HP SUM and components from the SDR using common YUM commands. For more information about downloading and installing HP SUM as an RPM, see the document *Linux best practices using HP Service Pack for ProLiant (SPP) and Software Delivery Repository (SDR)* at <http://h20000.www2.hp.com/bc/docs/support/SupportManual/c03479393/c03479393.pdf>.

For information on using the SDR, go to <http://www.hp.com/go/sdr>.

5 Advanced topics

Configuring IPv6 networks

You can deploy to remote nodes in IPv6-based networks for Windows and Linux node servers. Using HP SUM with IPv6 networks presents challenges for IT administrators.

For Windows-based servers, to communicate with remote node servers, HP SUM uses either existing credentials or the user-provided user name and password to connect to the admin\$ share. This share is an automatic share provided by Windows Server. After HP SUM connects to the admin\$ share, it copies a small service to the node server for the duration of the installation. After this service starts, HP SUM uses this service to communicate between the local and remote node server. During this process, HP SUM opens ports in the Windows firewall to enable HP SUM to communicate with the HP SUM engine on the node over SSL to pass data among local and remote systems. For more information about the ports used, see [“Network ports used by HP SUM” \(page 63\)](#). After the installation is completed or canceled, HP SUM stops the remote service, removes it from the node, closes the port on the Windows firewall, and then releases the share to the node server admin\$ share.

For Linux-based servers, to communicate to remote node servers, HP SUM starts by using the user-provided user name and password to create a SSH connection to the node server. After it connects, HP SUM copies a small service to the node server for the duration of the installation. After this service starts, HP SUM uses this service to communicate between the local and remote node server. During this process, HP SUM opens ports in the iptables firewall to enable HP SUM to communicate with the HP SUM engine over SSL to pass data between the local and remote systems. For more information about the ports used, see [“Network ports used by HP SUM” \(page 63\)](#). When the installation is completed or canceled, HP SUM stops the remote service, removes it from the target server, closes the port in the iptables firewall, and then closes the SSH connection to the node server.

To set up IPv6 networking, refer the documentation for your operating system.

Network ports used by HP SUM

HP SUM requires that certain network ports are available for proper operation. If you lock down network ports, make sure that the ports listed in the network port tables are open so that HP SUM works correctly when connecting to remote node servers and hosts. If you are unable to unlock these network ports, the only option is to run HP SUM locally and update network-based hosts, such as the OA, iLO, and VC modules, through their web interfaces.

Updates for most node types require network traffic in both directions between the server running HP SUM and the node. The server running HP SUM creates a local HTTP server, which is used to serve firmware binaries to the node and to communicate node status. The remote node issues HTTP requests and posts status updates to the server running HP SUM during the update process. If there is a routing problem or firewall blocking traffic back from the remote node to the system running HP SUM, firmware updates might be blocked, status updates blocked or delayed, or both.

Table 2 HP SUM Windows network ports

Ports	Description
Port 22	Establishes a connection to a remote node via SSH to perform node inventory.
Port 443	A secure data port used to transfer information.
Ports 445 and 137/138/139 (Port 137 is used only if you are using NetBIOS naming service.)	Connects to the remote ADMIN\$ share on node servers. These are the standard ports Windows servers use to connect to the remote file shares. If you can connect remotely to a remote Windows file share on the node server, you have the correct ports open.

Table 2 HP SUM Windows network ports *(continued)*

Ports	Description
Port 62286	Default for some internal communications. This port listens on the remote side if there is no conflict. If a conflict occurs, the next available port is used.
Ports 63001–63002	Updates are passed to the node and retrieved through an internal secure web server that uses the first available port in the range of 63001-63002. This allows iLO and VC firmware updates without having to access the host server. It also allows the servers to run VMware or other virtualization platforms to update the iLO firmware without requiring a server reboot or a migration of the virtual machines to other servers. Remote HP Integrity iLO and Superdome 2 updates require these ports to be open on systems for network traffic in both directions to transfer firmware files.
Ports 21 or 63006–63010	You can use these FTP ports to perform switch updates.

Table 3 HP SUM Linux network ports

Ports	Description
Port 22	Establishes a connection to a remote node via SSH to perform node inventory.
Port 443	A secure data port used to transfer information.
Port 62286	Default for some internal communications. This port listens on the remote side if there is no conflict. If a conflict occurs, the next available port is used.
Ports 63001–63002	Updates are passed to the node and retrieved through an internal secure web server that uses the first available port in the range of 63001-63002. This support allows iLO and VC firmware updates without having to access the host server. It also allows servers running VMware or other virtualization platforms to update their iLO without having to reboot their server or to migrate their virtual machines to other servers. Remote HP Integrity iLO and Superdome 2 updates require these ports to be open on systems for network traffic in both directions to transfer firmware files.
Ports 21 or 63006–63010	You can use these FTP ports to perform switch updates.

Special network configuration note for HP Integrity servers

HP Integrity servers have management network and production interfaces. These are usually kept on separate subnets in an installation. To perform full remote administration of the server, access is required for both networks. If you keep both networks isolated, you need to perform management and operating systems tasks separately.

GatherLogs

HP SUM provides a tool that collects all log files into one file. If you are troubleshooting an issue, run `GatherLogs`. The script is available in the same directory that holds HP SUM.

6 Troubleshooting

Collecting trace directories

HP SUM generates a set of debug trace logs located in the %TEMP%\HPSUM directory on Windows systems. These files contain internal process and debug information which can be useful in determining HP SUM failures.

The debug trace files are located under %temp%\hp_sum for Windows. The log files are located under C:\cpqsystem\hp\log. These files provide the following information and are appended in each HP SUM session.

HP SUM 5.0.0 and later includes a utility named GatherLogs.bat (Windows) or Gatherlogs.sh (Linux) to create a compressed .zip (Windows) or tar.Z (Linux) file with all the logs. If you need to review the log files, you can run this utility to gather all the logs in one file.

NOTE: Exit HP SUM before running the GatherLogs utility.

Debug Trace Files	Function
InventoryResults.xml	Contains details of the component inventory from the repositories.
Settings.xml	Includes general settings information of HP SUM such as Force downgrade or upgrade.
SourceClient.trace	Includes trace data of repository manager and general errors if any.
Hpsumiserver\Hpsumiserver.log	Contains trace data for HP SUM SOAP server sessions.
Hpsumiserver\HpsumserverW32.log	Contains remote trace data for HP SUM SOAP server sessions.
Hpsumiserver\localhpsumsoapserver.log	Contains information of the HP SUM SOAP server.
Sesssion.log	Contains the data and time for each session has started. This file is saved in separate directory named with the date.
RepositoryManager	Provides the repository and component information This directory can be excluded in the trace data when collecting the trace files.
<target>\Discoverymanager.log	Provides the details of interaction between the Operations Manager and the remote discovery client. If a discovery tool fails, it is reported to this trace file and surfaced as a Discovery Failed message. This log is target specific.
<target>\Installmanager.log	Provides the interaction between the Operations Manager and the remote discovery client. If a discovery tool fails, it is reported to this trace file and surfaced as a Discovery Failed message. This log is target specific.
<target>\<target name>_log.txt	Provides the trace data from operations manager for specific target. <target> is the name of the target in the source selections screen.
<target>\Settings.xml	Provides general settings information of HP SUM such as Force downgrade or upgrade for specific target.

Log Files	Function
<target>\hpsum_log.txt	Contains information of HP SUM discovery, installation status and errors if any. <target> is the name of the target in the source selections screen.
Hpsum_detail_log.txt	Contains the log data of the components.
hpsum.ini	Stores persistent data on the user's system.

If you are running HP SUM in offline mode, use the following instructions to collect trace directories and logs.

1. Launch HP SUM in offline mode.
2. Launch the command prompt from the HP SUM GUI by pressing **CTRL-ALT-D-B-X**.

NOTE: After approximately 30 seconds, the command prompt will appear over the HP SUM GUI window.

3. Change the directory to the directory running HP SUM. For example, `cd /mnt/bootdevice/SPP2012060B/hp/swpackages`.
4. Type `./GatherLogs.sh` to collect the HP SUM logs. All logs are collected in a `.tar.gz` file in the directory where you placed HP SUM. The log file is named `HPSUM_Logs_$(datetime).tar`.
5. Place the logs on a removable media if you want to view them on another computer.

Baseline troubleshooting

HP SUM lists SUSE Enterprise Linux dependencies for Red Hat Enterprise Linux systems

Solution: No action to take. HP SUM incorrectly lists the SUSE components. HP SUM will install the Red Hat components.

HP SUM displays two versions of the same component when creating a custom baseline

Solution: Multiple instances of the same component can be seen when creating a custom baseline because the same component exists in the multiple baselines that were selected to be included in the custom baseline. When you create a custom baseline, make sure you only select one version of a component if it is included in multiple source baselines.

HP SUM stops responding when performing inventory on large baselines

Solution: Only perform inventory on one large (SPP-size) baseline at a time. Performing inventory on more than one inventory might use too many system resources.

HP SUM does not return an error when you enter incorrect credentials for **Download from hp.com**

Solution: Verify that the user credentials are correct.

Node troubleshooting

HP SUM does not respond when editing a node with `sudo`

Solution: Make sure the user that is logged into HP SUM is part of the administrator group for the system.

HP SUM does not display the **Deploy** button on the Deploy Nodes screen

Solution: Make sure that HP SUM has resolved all dependencies before attempting to deploy updates. If HP SUM discovers a failed dependency, it does not activate the **Deploy** button.

An HP-UX node displays the error Inventory failed. The pciinfo module requires manual update on remote target

Solution: Use the following instructions to resolve the error:

1. Close all tasks and back up important data on the node server.
2. Log in to the node server with root privileges.
3. Transfer the file `pciinfo.depot`, located in the HP SUM directory `/ia64/` to the node server.
4. Run `swinstall -x autoreboot=true -s 'pwd' ./pciinfo.depot PCCIINFO`.
5. After the update finishes, you can re-run HP SUM on a remote host to continue with additional updates.

HP SUM does not detect solid state hard drives

Solution: If HP SUM does not detect the solid state hard drive, run the smart component individually.

After HP SUM finishes an inventory, the inventory screen does not close

Solution: Click the **Cancel** button to close the inventory screen.

Cannot add a Windows node when running HP SUM on a Linux host

Solution: This is a known issue. Run HP SUM on a Windows host to add Windows nodes.

HP SUM does not display the **Reboot** button after it finishes a deployment

Solution: Verify that no node updates had issues. HP SUM does not display the **Reboot** button until all updates are successful. Do not reboot the node until all issues are resolved. This minimizes the chances that components can be on unsupported versions.

HP SUM does not automatically reboot a node

Solution: Verify all components updated properly. If there are no issues, reboot the node manually.

HP SUM does not display a reboot message in Guided Update

Solution: No action to take.

When multiple nodes are selected, HP SUM does not display the most correct status

Solution: No action to take. This is a known issue. To view the correct state, view a single node.

HP SUM displays message Unable to locate the item you requested

Solution: Press **F5** when HP SUM displays this message.

HP SUM stops responding when performing inventory on multiple nodes

Solution: Inventory only one node at a time. When performing inventory on multiple nodes at the same time, the host might run low on resources.

HP SUM displays the message Unable to login or identify node as a supported device when HP SUM cannot find a node

Solution: Ping the node to make sure it is on the network. HP SUM might not be able to connect to the node.

HP SUM displays the message Ready for deployment while HP SUM is performing inventory on a node

Solution: Make sure that HP SUM completes an inventory before attempting to deploy updates to a node.

HP SUM does not display an entire screen

Solution: Change the resolution of a display or browser.

HP SUM displays extra credentials boxes on the Edit Node screen

Solution: Do not use the boxes beneath the **Access Level** section.

You can edit a node while HP SUM is deploying updates to the node

Solution: Wait for HP SUM to finish deployment before editing a node. Editing a node before HP SUM finishes deployment might cause issues with the deployment.

HP SUM displays an associated node multiple times

Solution: No action to take. HP SUM lists the associated node each time that it performs inventory based on the main node.

HP SUM automatically performs inventory on a node after editing the node

Solution: No action. This is a known issue.

Activity screen troubleshooting

HP SUM does not update the Activity screen accurately

Solution: No action. This is a known issue.

VMware troubleshooting

HP SUM displays an inventory error on a VMware node

Solution: Use the following steps to make sure the VMware service is running.

1. Log in to the VMware ESXi shell.
2. Check the provider version, type `esxcli software vib list | grep smx`.
3. Verify that the provider service is running, type `/etc/init.d/sfcbd-watchdog status`.
4. Verify that the provider responds to queries, type `enum_instances SMX_SCInstallationService root/hpq`.
5. If you need to restart the service, type `/etc/init.d/sfcbd-watchdog restart; /etc/init.d/sfcbd-watchdog status;`

Reports troubleshooting

HP SUM does not generate reports for nodes that are offline

Solution: Verify that HP SUM can ping the node.

HP SUM displays only requested reports in CSV and XML format

Solution: No action to take. This is the expected result.

HP SUM service troubleshooting

HP SUM service stops running when HP SUM is mounted from a virtual media source

Solution: Do not mount a virtual media source through an iLO. A known issue causes the iLO to restart and causes HP SUM to stop working.

HP SUM allows all users to log in

Solution: All users can log into HP SUM. Only users with administrator permission can deploy updates or add nodes.

Some screens are not translated from English

Solution: No action. This is a known issue that will be fixed in a future version.

Legacy CLI commands troubleshooting

HP SUM does not deploy multiple components at in one command

Solution: Copy the components into a directory and use the `/use_location` command. If you are deploying more than ten components, use the work around.

The `/use_web` Legacy CLI command does not work

Solution: No action. This is a known issue. This version of HP SUM does not support this command.

7 Support and other resources

Contacting HP

For worldwide technical support information, see the HP support website:

<http://www.hp.com/support>

Before contacting HP, collect the following information:

- Product model names and numbers
- Technical support registration number (if applicable)
- Product serial numbers
- Error messages
- Operating system type and revision level
- Detailed questions

Subscription service

HP recommends that you register your product at the Subscriber's Choice for Business website:

http://www.hp.com/country/us/en/contact_us.html

After registering, you will receive email notification of product enhancements, new driver versions, firmware updates, and other product resources.

Related information

Documents

You can find these documents on the SPP documentation page:

<http://www.hp.com/go/spp/documentation>

- *HP Smart Update Manager Release Notes*
- *HP Service Pack for ProLiant Release Notes*
- *HP Service Pack for ProLiant Server Support Guide*
- *HP ProLiant and Integrity Firmware Management Best Practices Overview*
- *HP ProLiant and Integrity Firmware Management Best Practices Planning Guide*
- *HP ProLiant and Integrity Firmware Management Best Practices Implementer Guide*

The following document is available for HP Integrity firmware update best practices:

- *Using HP Smart Update Manager with Integrity Servers*
<http://bizsupport2.austin.hp.com/bc/docs/support/SupportManual/c03572337/c03572337.pdf>

Websites

For more information about HP SUM, see the HP SUM website:

<http://www.hp.com/go/hpsum>

For more information about SPP, see the SPP website:

<http://www.hp.com/go/spp>

To download the latest SPP and available hot fixes, see the SPP download page:

<http://www.hp.com/go/spp/download>

For information about HP Subscriber's Choice, see the Subscriber's Choice website:

<http://www.hp.com/go/subscriberschoice>

For information on the HP Systems Insight Manager, see the following documents on the HP Systems Insight Manager website:

<http://www.hp.com/go/insightmanagement/sim/docs>

- *HP Systems Insight Manager Installation and Configuration Guide for MS Windows*
- *HP Systems Insight Manager User Guide*

For more information about HP ProLiant Gen8 servers and software see the HP website:

<http://www.hp.com/go/proliantgen8/docs>

For information about HP Insight Control Management Software, see the HP website:

<http://www.hp.com/servers/rdp>

For information about operating systems supported by HP ProLiant servers, see the operating system support matrices:

<http://www.hp.com/go/supportos>

For information about support for updating SATA hard drives in a Modular Smart Array 20/50/60/70 storage enclosure connected to an HP ProLiant server using a Smart Array controller, see the support matrix on the HP StorageWorks Modular Smart Arrays website:

<http://www.hp.com/go/msa>

Typographic conventions

Table 4 Document conventions

Convention	Element
Blue text: Table 4 (page 71)	Cross-reference links and e-mail addresses
Blue, underlined text: http://www.hp.com	Website addresses
Bold text	• Keys that are pressed • Text typed into a GUI element, such as a box • GUI elements that are clicked or selected, such as menu and list items, buttons, tabs, and check boxes
<i>Italic</i> text	Text emphasis
Monospace text	• File and directory names • System output • Code • Commands, their arguments, and argument values
<i>Monospace, italic</i> text	• Code variables • Command variables
Monospace, bold text	Emphasized monospace text



CAUTION: Indicates that failure to follow directions could result in damage to equipment or data.

❗ **IMPORTANT:** Provides clarifying information or specific instructions.

NOTE: Provides additional information.

💡 **TIP:** Provides helpful hints and shortcuts.

HP Insight Remote Support software

HP strongly recommends that you install HP Insight Remote Support software to complete the installation or upgrade of your product and to enable enhanced delivery of your HP Warranty, HP Care Pack Service, or HP contractual support agreement. HP Insight Remote Support supplements your monitoring continuously to ensure maximum system availability by providing intelligent event diagnosis, and automatic, secure submission of hardware event notifications to HP, which will initiate a fast and accurate resolution, based on your product's service level. Notifications may be sent to your authorized HP Channel Partner for onsite service, if configured and available in your country.

For more information, see the HP website at <http://www.hp.com/go/insightremotesupport>. The *HP Insight Remote Support Release Notes* detail the prerequisites, supported hardware, and associated operating systems. The release notes are available on the HP website at <http://www.hp.com/go/insightremotesupport/docs>. HP Insight Remote Support is included as part of HP Warranty, HP Care Pack Service, or HP contractual support agreement.

HP Insight Online

HP Insight Online is a new capability of the HP Support Center portal. Combined with HP Insight Remote Support 7.x, it automatically aggregates device health, asset, and support information from iLO Management Engine with contract and warranty information, and then secures it in a single, personalized dashboard that is viewable from anywhere at any time. The dashboard organizes your IT and service data to help you understand and respond to that information more quickly. With specific authorization from you, an authorized HP Channel Partner can also remotely view your IT environment at HP Insight Online.

- For more information about using HP Insight Online, see the *HP Insight Online Getting Started Guide* at <http://www.hp.com/go/proliantgen8/docs>.
- To install HP Insight Remote Support and enable HP Insight Online, see the *HP Insight Online Integrated Solution and Management Setup Guide* at <http://www.hp.com/go/proliantgen8/docs>.

8 Documentation feedback

HP is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback@hp.com). Include the document title and part number, version number, or the URL when submitting your feedback.

Glossary

AMS	Agentless Management Service
Analyze	The process where HP SUM verifies all dependencies before deploying updates
Baseline	A set of updates for your hardware
CPLD	complex programmable logic device
Deploy	The process where HP SUM begins the update process by installing the component update files to the node.
DNS	domain name system
Guided Update	An automated or wizard-style method to update the local node
iLO	Integrated Lights-Out
Inventory	The process where HP SUM determines baseline and node contents
iPDU	intelligent power distribution unit
ISP	Integrity Support Pack
Location	Network directory for baseline updates
Node	A piece of hardware you want to update, for example a server, iLO, or NIC.
OA	Onboard Administrator
RILOE II	Remote Insight Lights-Out Edition II
Scouting	The process where HP SUM determines node type
SDR	Software Delivery Repository
SPP	HP Service Pack for ProLiant
SSH	Secure Shell
SSL	Secure Sockets Layer
TPM	Trusted Platform Module
UNC	Universal Naming Convention
URI	Uniform resource identifier, commonly known as the web address
VC	Virtual Connect
WBEM	Web-Based Enterprise Management
WMI	Windows Management Instrumentation

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